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Study Report

STRATEGIC MAPPING OF THE EDTECH ECOSYSTEM IN AFRICA

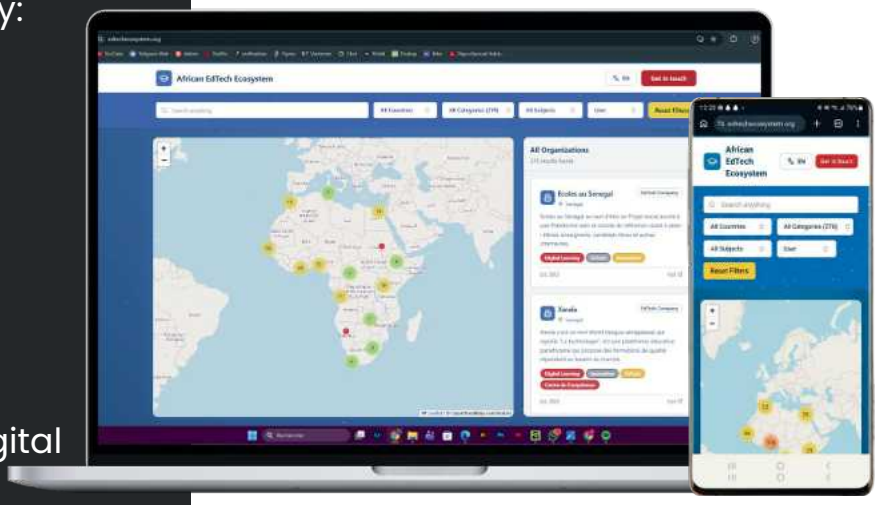
Dakar EdTech Hub / timbuktoo Program

STRATEGIC MAPPING OF THE EDTECH ECOSYSTEM IN AFRICA

This study was commissioned by the UNDP's timbuktoo Programme as part of the creation of the Dakar EdTech Hub, with the objective of structuring, analyzing and energizing the African educational technology ecosystem.

The research, mapping and strategic analysis mission was conducted by:

Abdou Khadre POUYE
Senior Consultant - Ecosystems, EdTech & Digital Innovation



To ensure the sustainability of the mapping initiative, a dedicated digital platform has been established:

https://edtechecosystem.org/

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List of abbreviations and acronyms

AI – Artificial Intelligence
AU – African Union
AVU – African Virtual University
AfDB – African Development Bank
B2B – Business to Business
B2G – Business to Government
CBC – Competency-Based Curriculum
ECOWAS – Economic Community of West African States
DLP – Digital Literacy Programme
EdTech – Education Technology
EMIS – Education Management Information System
ICT – Information and Communication Technologies / Information and Communication Technologies
ICT4E – Information and Communication Technologies for Education
KPIs – Key Performance Indicators
K-12 – Kindergarten to Grade 12
LMS – Learning Management System
MCF – Mastercard Foundation
MOOC – Massive Open Online Course
MVP – Minimum Viable Product
NEMIS – National Education Management Information System (Kenya)
OER – Open Educational Resources
PPP – Public-Private Partnership
PSEF – Education and Training Sector Program
REB – Rwanda Education Board
Series A / Series B – Mid-Round and Growth Funding Rounds
SIMEN – National Education Management Information System (Senegal)
SIS – Student Information System (Academic Information System)
SN2025 – Digital Senegal Strategy 2025
STEAM – Science, Technology, Engineering, Arts and Mathematics (Sciences, technologies, engineering, arts and mathematics)
TVET – Technical and Vocational Education and Training;
UA – African Union;
UNCHK – Cheikh Hamidou Kane Digital University;
UNDP – United Nations Development Program;
UVS – Virtual University of Senegal;
VC – Venture Capital;
WAEMU – West African Economic and Monetary Union

THE TIMBUKTOO PROGRAM

A STRUCTURING LOGIC AT THE SERVICE OF AFRICAN INNOVATION

The timbuktoo program, led by the United Nations Development Programme (UNDP), is a new-generation pan-African initiative aimed at fundamentally transforming the innovation ecosystem in Africa. Unlike traditional approaches focused on isolated projects or ad hoc support for startups, timbuktoo adopts a systemic approach, geared towards the sustainable structuring of strategic value chains with strong economic and social impact.

A response to the limitations of traditional models for supporting innovation:

The UNDP analysis highlighted several structural limitations of African innovation ecosystems: fragmented actors, weak patient funding, difficulty accessing public procurement, lack of alignment with national policies, and low production of impact evidence. timbuktoo was designed to address these challenges by combining innovation, public policy, and structural funding within a single operational framework.

The foundations of the timbuktoo program

The program is based on three structuring principles:

1. timbuktoo's strategic sector approach targets sectors with high potential for transformation and societal impact, including EdTech, Agritech, HealthTech and ClimateTech, in order to concentrate resources, structure value chains and foster the emergence of African champions.

2. Deployment through specialized hubs Each sector is supported by a regional hub, anchored in a dynamic national ecosystem and connected to a pan-African network of public, private and academic partners.

3. With catalytic capital and a strong institutional anchor, timbuktoo mobilizes patient financial resources and works closely with states to align innovative solutions with national development priorities and public policies.

Why is EdTech a priority sector for timbuktoo?

Digital education (EdTech) occupies a central place in timbuktoo's vision, due to its cross-cutting role in skills development, youth employability, and the resilience of education systems. Faced with demographic pressure, inequalities in access, and the limitations of traditional educational models, EdTech emerges as a strategic lever for systemic transformation, capable of operating on a large scale while adapting to local contexts.

However, the African EdTech ecosystem remains fragmented, undercapitalized, and insufficiently integrated into public education systems. timbuktoo's intervention aims precisely to bridge this gap between local innovation, educational policies, and structuring funding mechanisms.

FROM THE TIMBUKTOO PROGRAM TO THE ROADMAP OF THE DAKAR EDTECH HUB

The creation of the Dakar EdTech Hub is part of this sectoral structuring strategy. The Hub represents the operational implementation of timbuktoo's vision for EdTech in Francophone Africa. It aims to serve as a regional platform, linking innovation, regulation, and financing.

The roadmap for the Dakar EdTech Hub stems directly from this approach and is organized around four complementary axes:

1. Structuring the EdTech ecosystem: Mapping the players, uniting initiatives, strengthening regional synergies and reducing market fragmentation.

2. Supporting the growth and scaling of EdTech solutions. Deploying specialized incubation and acceleration programs, facilitating access to patient capital and supporting scaling.

3. Create an interface between innovation and public policy. Set up a Policy Lab enabling the co-construction of regulatory frameworks, the experimentation of solutions and the integration of EdTech into national education systems.

4. Produce and promote evidence of impact. Develop an Evidence Lab to measure the effectiveness of solutions, document results and inform public decisions and investments.

A coherent trajectory towards systemic transformation

Thus, timbuktoo is not an isolated initiative, but the strategic foundation from which a clear operational roadmap for EdTech is being developed. The Dakar EdTech Hub is the implementation tool, with the ambition of catalyzing a sustainable transformation of African education systems, serving inclusion, employability, and human development.



SECTION 1

GENERAL INTRODUCTION

1. DEFINITION AND CONTEXT OF EDTECH IN AFRICA

EdTech, or technology applied to education, encompasses a range of learning platforms, digital tools, interactive content, open educational resources (OER), mobile technologies, artificial intelligence solutions, and infrastructure to support teaching and learning. In Africa, this sector is emerging as a strategic response to the structural challenges of education systems and as a lever for digital transformation, skills development, and socio-economic inclusion.

The African continent is at a pivotal moment in its development. With nearly 60% of its population under the age of 25[1], Africa represents the world's largest concentration of young people. By 2050, it will be home to nearly half of the global workforce[2]. This demographic dynamic is placing considerable pressure on education systems, which are called upon to provide massive, rapid, and context-specific training.

At the same time, the technological landscape is changing: more than 600 Tech hubs[3] are driving innovation ecosystems in major hubs such as Cape Town, Lagos, Nairobi, Cairo or Dakar.

However, these ecosystems often remain fragmented, undercapitalized, and insufficiently interconnected, limiting their capacity to generate a large-scale systemic impact. Educational challenges are numerous and profound: school dropout rates among the highest in the world, marked territorial inequalities, poor connectivity, a lack of digital infrastructure, a shortage of teachers trained in the use of technology, and a scarcity of contextualized educational content, particularly in local languages.

Given this situation, EdTech is a key lever for improving access to, the quality of, and the relevance of learning. It expands educational opportunities, modernizes the governance of education systems, strengthens youth employability, and supports the transition to an inclusive digital economy.

It is in this perspective that the UNDP's timbuktoo initiative is situated, which positions Dakar as a pan-African EdTech Hub intended to unite, structure and catalyze educational innovation dynamics across the continent.

1. UNFPA – State of World Population Report (2023)
2. World Bank – Global Monitoring Report 2022
3. GSMA – Africa's Tech Landscape Report (2023)

2. REPORT METHODOLOGY AND DATA SOURCES

The methodology used for this report is based on a rigorous, multi-source analytical approach, structured around the following axes:

(1) In-depth documentary analysis

The report is based on a comprehensive review of strategic documents, benchmark studies, and recent publications, including:

- timbuktoo UNDP Policy Lab – Startup policy frameworks
- Injini Africa EdTech Insights Reports (2023–2024)
- Rapid Scan EdTech – Senegal (EdTech Hub, 2020)
- WAEMU EdTech Study (2023)
- ICT in Secondary Education in Sub-Saharan Africa (2019)
- 10 Actions to Strengthen Africa's EdTech Ecosystem – Mastercard Foundation (2024)
- Tertiary Education and Skills (TES) Multi-Donor Trust Fund (World Bank Group)
- Education Data Mapping in Sub-Saharan Africa (Unicef)
- The EdTech landscape in Rwanda (2023)
- Mapping education data in sub-Saharan Africa (2020)

(2) Qualitative field interviews

Interviews were conducted with startup founders, technology hubs, public institutions, and technical partners. Data was collected via standardized forms (Company Interview Form, HUB Form, Program Form), notably within the framework of the Mastercard Foundation's EdTech Fellowship program.



(3) Regional and continental strategic reading

The analysis is grounded in the following strategic frameworks:

- Vision Senegal 2050 – New Technological Deal
- Digital Senegal Strategy 2025 (SN2025)
- African Union Agenda 2063
- Strategy for Digital Education in Africa (2023–2028)
- Global SDG4 – Education 2030 Framework for Action
- A 16-25)
- ECOWAS ICT in Education Regional Framework
-
- Rwanda Education Sector Strategic Plan
- Kenya National ICT Master Plan
- South Africa, White Paper on e-Education
- Ghana, ICT in Education Policy, Curriculum reforms (CBC)
- Nigeria, National Digital Economy Policy

(4) Comparative and systemic approach

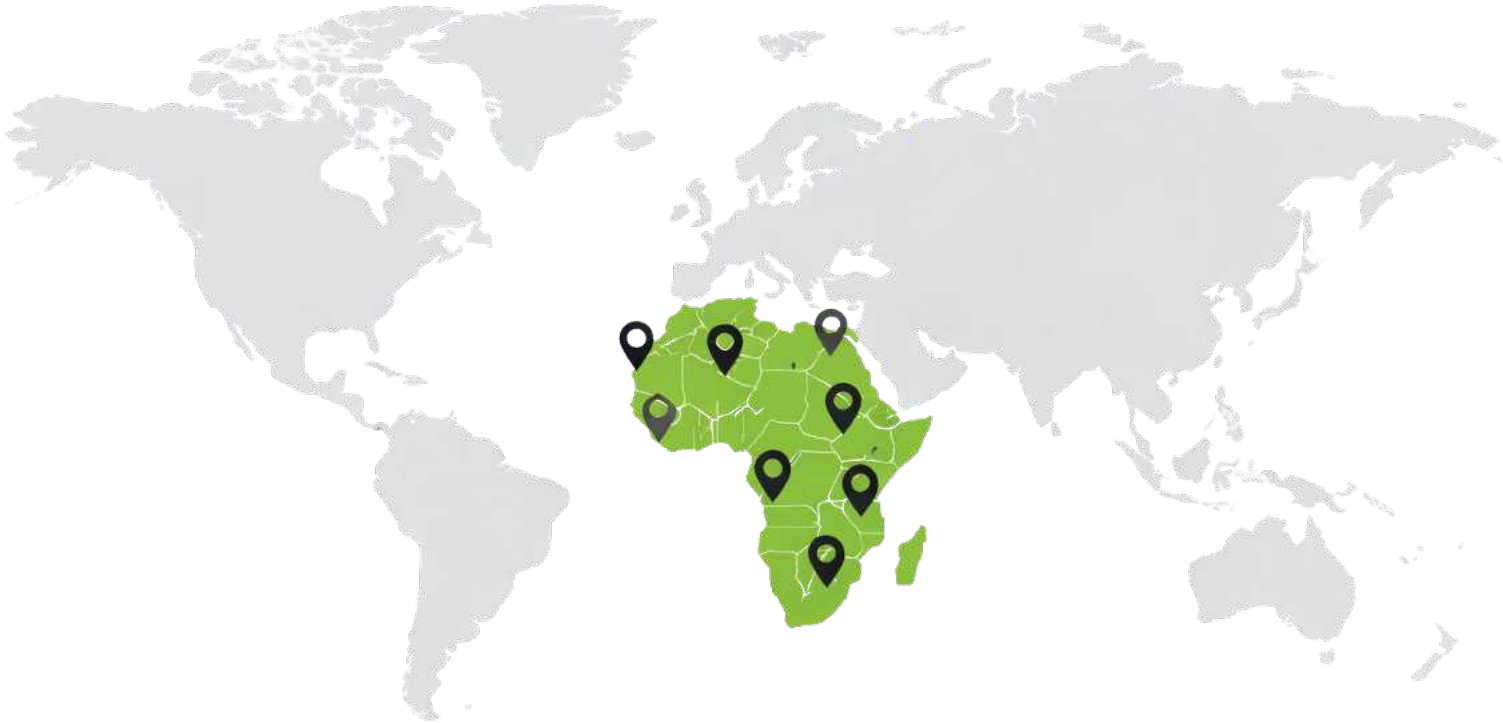
The methodological approach is based on:

- A mapping of the actors in the EdTech ecosystem (sectors, geographical areas, intervention models)
- A pedagogical classification based on the HolonIQ, Injini and EdTech Hub frameworks
- A cross-analysis of technological, economic, and institutional models
- International benchmarks adapted to African realities (countries comparable in terms of infrastructural, demographic and institutional development)



3. OBJECTIVES & STRUCTURE OF THE REPORT

THIS REPORT PURSUES FOUR MAIN OBJECTIVES:



ANALYZING THE AFRICAN EDTECH ECOSYSTEM

Understanding the key dynamics

- Map the actors (startups, hubs, governments, funders).
- Identify the major obstacles: funding, institutional adoption, infrastructure.
- Identify the levers for acceleration: local innovation, public policies, partnerships

DEVELOP A PAN-AFRICAN STRATEGIC MAP

Multi-criteria segmentation:

- By educational levels: K-12, higher education, TVET/vocational training.
- By geographical area: West Africa, East Africa, Southern Africa, North Africa
- By type: content, LMS platforms, data & analytics, hardware, inclusion, educational AI, etc.

DOCUMENTING TECH HUBS AND STRUCTURING PROGRAMS

Analyze the support structures

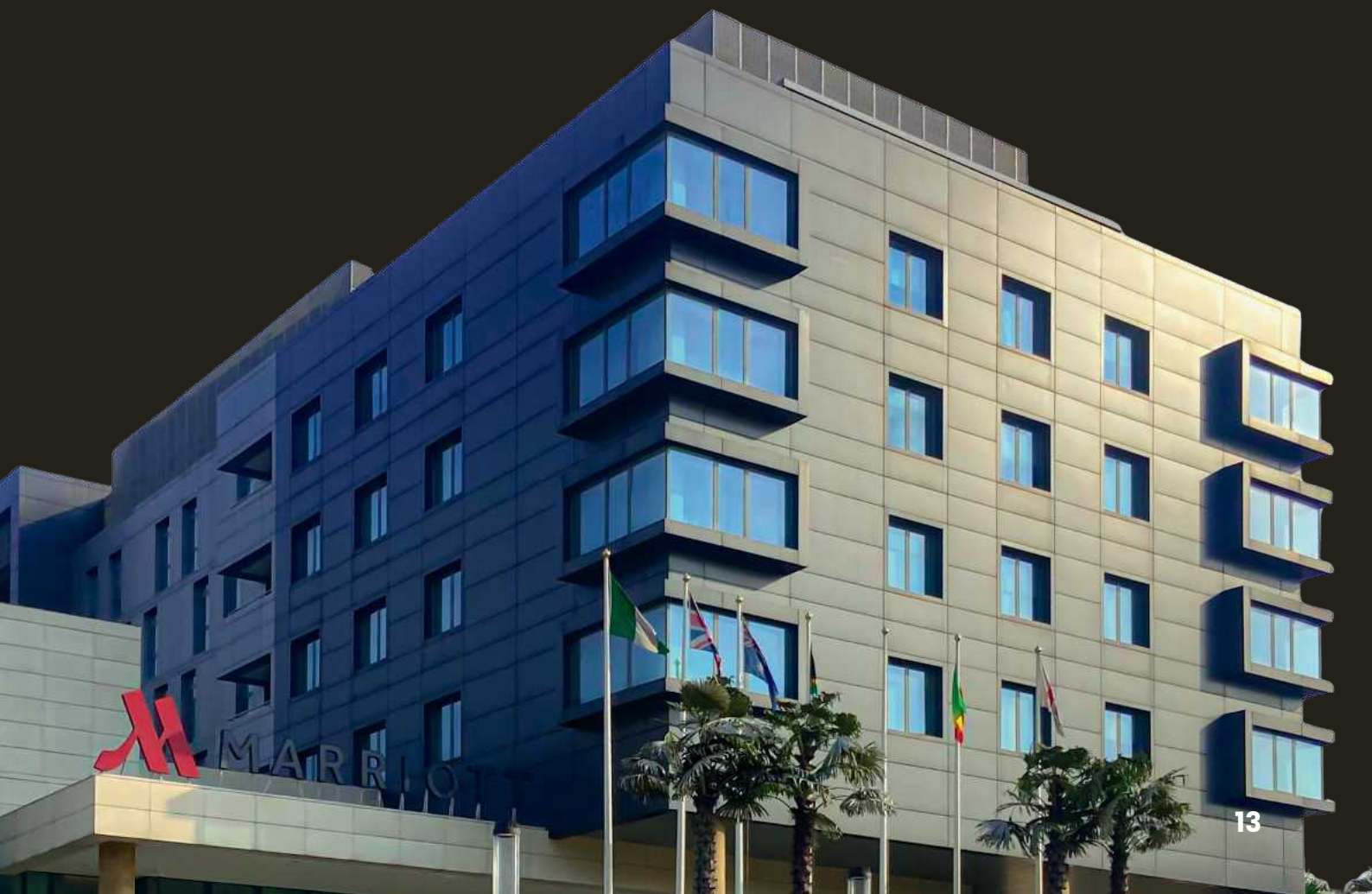
- Identify high-impact hubs (acceleration, incubation, fablabs, universities).
- Highlighting regional programs
- Identify the enabling conditions: funding, talent, policies, infrastructure.

CO-CREATING THE ROADMAP FOR THE DAKAR EDTECH HUB

Strategic alignment

Operational model inspired by the timbuktoo – UNDP framework.

Consistency with national priorities: digital transformation, inclusion, skills of the future.
Addressing the needs of the French-speaking EdTech market: scalability, proof of impact, interoperability, applied research.
Definition of deliverables: hub services, startup pipeline, funding mechanisms, university partnerships, impact indicators.



SECTION 3

DETAILED MAPPING OF EDTECH COMPANIES & MARKET DYNAMICS

1. GENERAL OVERVIEW AND DYNAMICS OF THE AFRICAN EDTECH MARKET

The African EdTech ecosystem is evolving at the intersection of three major dynamics.



DEMOGRAPHIC PRESSURE
UNPUBLISHED



CHANGING NEEDS
EDUCATIONAL



INCREASE IN POWER
TECHNOLOGICAL INNOVATION

Despite significant potential, the sector remains limited by structural constraints that slow its maturation and scaling up.

A. Very strong demographic growth

Africa is experiencing a phase of demographic growth unique in the world, marked by the rapid expansion of its youth population. This evolution is creating a historic shift: the majority of the population is gradually entering school, university or professional age, leading to an unprecedented increase in demand for education, training and economic opportunities.

This pressure is being exerted on education systems already facing significant constraints: overcrowded infrastructure, a shortage of qualified teachers, limited training capacity, and persistent disparities between urban and rural areas. In many countries, demand now far exceeds traditional schooling capacity, forcing governments to rethink their educational delivery models.

This dynamic demands a profound transformation: training more young people, faster, and using approaches adapted to local realities and the aspirations of new generations. It also calls for a diversification of learning pathways, the modernization of teaching tools, and a better integration of education, digital skills, and economic opportunities.

In this context, the rise of a generation that is largely connected and receptive to digital uses makes technological innovation an essential catalyst to absorb this growing demand and lay the foundations for inclusive and sustainable development.

infrastructure saturation, lack of qualified teachers, limited training capacities and persistent disparities between urban and rural areas.

These factors reinforce the urgency of deploying EdTech solutions capable of operating at scale, while remaining hybrid and contextualized. They are part of a major demographic and territorial transformation: according to the Economic Report on Africa 2017 by the Economic Commission for Africa (ECA/UNECA), the share of the African population living in urban areas has increased from 31% in 1990 to a projected 49% in 2035.

This rapid urbanization is profoundly reconfiguring the structure of educational demand, giving rise to dynamic, heterogeneous and constantly expanding urban markets, including in peri-urban areas and informal settlements where the needs for access, flexibility and quality of learning are increasing.

B. A growing but underfunded market

Although the African EdTech ecosystem is experiencing sustained growth, it remains largely underfunded relative to its potential. Cross-estimated data from studies conducted by the West African Economic and Monetary Union (UEMOA), Injini, and EdTech Hub places the current size of the African market between USD 600 million and USD 1 billion, a very modest figure compared to the USD 300 billion global EdTech market.

Despite this low capitalization, the development prospects are particularly encouraging:

Projected annual growth of between 15% and 20% by 2035, supported by increasing demand for digital learning services.

The rise of flexible learning methods, including online training, micro-credentials, and short professional programs; mobile penetration exceeding 90% in several African countries, paving the way for "mobile-first" models[3]; more than 600 active tech hubs across the continent, forming a growing network for incubation and innovation[4]

The COVID-19 pandemic has also acted as an unprecedented accelerator. It has highlighted the limitations of traditional education systems and demonstrated the need for hybrid models, combining physical presence and digital learning. This period has also strengthened the acceptance of digital technology among institutions, teachers, and families, creating a window of opportunity for African EdTech.

However, this positive momentum is still hampered by several structural constraints: High market fragmentation, characterized by heterogeneous regulatory environments and low interoperability of solutions

Business models that are still fragile, with limited conversion and retention rates: The lack of suitable financing mechanisms, such as patient capital, seed funds or public guarantee schemes, which are nevertheless essential to support startups in their growth phase

[1] WAEMU EdTech Study, Mastercard Foundation & UEMOA (2024) - estimation of the EdTech market in Africa.
[2] Injini African EdTech Insights Report – Volume 2 (2024) EdTech business growth projections.
[3] EdTech Hub – Country Scans (Senegal, Kenya, Ghana, Rwanda, 2020–2023) — data on mobile penetration and digital usage.
[4] HolonIQ – Africa's Tech Hubs Landscape (2023) — census of more than 600 active innovation hubs on the continent.

To fully unleash the potential of EdTech on the continent, a coordinated effort between investors, public decision-makers and support structures is essential, particularly to structure a continuous funding chain from ideation to scaling up.

C. Persistent structural challenges

Despite growing dynamism and a proliferation of initiatives, the African EdTech ecosystem continues to face profound structural constraints of a financial, institutional, infrastructural, and human nature. These systemic obstacles limit the ability of innovations to achieve significant scale and produce a lasting impact on education systems.

1. Insufficient funding, ill-suited to the specific needs of EdTech

Funding is one of the sector's main bottlenecks. Less than 2% of technology venture capital invested in Africa is directed towards EdTech[1], placing this sector far behind fintech, e-commerce, or e-health. This under-allocation is reflected in particular by:

- a very limited rate of progression towards the Series A and B funding phases corresponding respectively to fundraising intended to support the scaling of the business model (Series A) and to accelerate growth, geographical expansion and organizational structuring (Series B) the majority of startups remaining confined to the pre-seed and seed stages[2];
- a high perception of risk, linked to dependence on public policies, slow budget cycles and the complexity of institutional B2G and B2B models;
- a lack of robust and standardized evidence of impact, particularly in terms of improving learning, teaching effectiveness or reducing educational inequalities[3].

2. Institutional adoption remains complex

The integration of educational technologies into public systems remains limited. Several persistent factors hinder this adoption:

- the complexity of regulatory frameworks and the slowness of accreditation or validation processes for digital content and solutions;
- the poor connectivity of schools, which restricts the daily educational use of digital platforms;
- the administrative overload of teachers, who are often poorly supported in adopting new teaching practices that integrate digital technology[4];
- the rigidity of curricular and pedagogical frameworks, leaving little room for experimentation, innovation and hybrid learning models.

These institutional constraints limit the systemic integration of EdTech solutions into national programs and hinder their large-scale deployment.

LESS THAN 2% OF AFRICAN TECHNOLOGY VENTURE CAPITAL IS DIRECTED TOWARDS EDTECH!

3. **Basic infrastructure still insufficient**

Educational and digital infrastructure is a major obstacle to the development of EdTech:

- less than 50% of primary schools in sub-Saharan Africa have reliable access to electricity[5];
- only 24% of households have access to the Internet[6], a rate significantly lower than the global average;
- marked structural differences between urban and rural areas, which accentuate the existing digital divide.

This lack of infrastructure complicates the use of online platforms, access to digital educational content and the equitable distribution of educational resources, particularly in rural and peri-urban areas.

4. **Limited human capabilities**

The digital transformation of education systems relies heavily on the skills of the stakeholders, which remain largely insufficient:

- a majority of teachers have not benefited from initial or continuing training in the use of educational technology and hybrid pedagogy[7];
- Education ministries rarely have specialized EdTech units capable of ensuring the strategic management, maintenance and supervision of digital projects;
- a low level of local ownership, linked to the absence of structured mechanisms for supporting change and continuing education.

Without a massive and coordinated strengthening of the skills of teachers, inspectors, technicians and administrators, EdTech initiatives remain fragile, dependent on external funding and difficult to sustain.

D. First signs of structuring of the African EdTech ecosystem

Despite a still heterogeneous level of maturity across regions and countries, several converging dynamics demonstrate the gradual entry of African EdTech into a structuring phase. These signals can be observed both at the continental level, through dedicated pan-African programs, and at the national level, via educational reforms explicitly integrating digital technology as a lever for systemic transformation.

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1 HolonIQ. Africa EdTech Market Outlook & Investment Landscape. HolonIQ, 2022–2024.
[2] Injini & Briter Bridges. African EdTech Investment Report. Injini, 2021 ; Briter Bridges, 2022.
[3] EdTech Hub. The Evidence Gap in EdTech: Why Impact Measurement Matters. EdTech Hub, 2020–2023.
[4] UNESCO. Teacher Readiness for Digital Learning in Sub-Saharan Africa. UNESCO, 2021.
[5] World Bank & UNESCO Institute for Statistics (UIS). Education Infrastructure and Access to Electricity in Sub-Saharan Africa. World Bank, 2022.
[6] International Telecommunication Union (ITU). Measuring Digital Development – Facts and Figures. ITU, 2023.
[7] UNESCO & UNICEF. Digital Learning and Skills Development in Africa. UNESCO–UNICEF, 2022.

1. The emergence of structuring continental programs

At the pan-African level, new mechanisms aim to move beyond fragmented approaches to sustainably structure strategic sectors of digital education.

The timbuktoo EdTech Hub (UNDP) is the first pan-African initiative explicitly designed to structure sectoral value chains, including EdTech, through a network of specialized regional hubs. Its aim is to catalyze innovation, strengthen access to finance, and integrate EdTech solutions into national development priorities.

The Mastercard Foundation EdTech Fellowship is part of a complementary approach to accelerating African EdTech startups, relying on partner Tech Hubs to provide integrated support combining funding, strategic coaching, production of evidence of impact and international networking[8].

These programs help to create bridges between local innovation, public policies and structuring financing, while promoting the emergence of a common foundation of standards and practices in the African EdTech ecosystem.

2. National educational reforms integrating digital technology

Beyond a few pioneering countries like Kenya, Rwanda, and Senegal, alongside continental initiatives, a growing number of African countries have undertaken national education reforms that place digital technology at the heart of modernizing learning systems. This dynamic is now spreading across the continent, with varying levels of ambition and implementation.

These reforms generally cover several key areas:

- the integration of digital skills into national curricula;
- the digitization of educational content and resources;
- the deployment of educational information systems (EMIS, LMS, SIS)[1] to improve governance and management;
- the promotion of hybrid teaching models, combining face-to-face and online learning;
- capacity building for teachers and administrative staff in the use of digital technology.

[1] EMIS – Education Management Information System- LMS – Learning Management System-SIS – Student Information System

Taken together, these developments reflect a growing recognition of digital education as a strategic lever to improve access, quality and resilience of African education systems, while opening up a space of opportunities for local and regional EdTech solutions.

digital infrastructure (connectivity, equipment)	**continuing education for teachers**
the creation of national educational portals	**Curriculum revision integrating digital skills and active learning methods**
the adoption of EdTech tools in school management (EMIS/SIMEN)	**distance learning and hybrid learning paths.**



BELOW IS A COMPREHENSIVE AND STRUCTURED COUNTRY-BY-COUNTRY PRESENTATION.

TABLE OF NATIONAL REFORMS FOR THE INTEGRATION OF DIGITAL TECHNOLOGY IN EDUCATION IN AFRICA

Country	Reform / Policy	Main areas and key actions
Egypt	Vision 2030 – Education Reform 2.0	• Egypt Knowledge Bank • 100% digital exams for secondary school
Sudan	Digital Education Initiative	• ICT integration despite instability • E-learning for displaced populations
Morocco	Morocco Digital 2020 / GENIE Program	• Equipment + teacher training • Telmig TICE & ACCADEMIA Platforms • Connectivity >95% of middle/high schools
Kenya	Competency-Based Curriculum (CBC), Digital Literacy Programme (DLP)	• Introduction of digital skills • >1 million tablets distributed • Hybridization of teaching methods • • national EMIS : NEMIS
Rwanda	ICT in Education Policy (2016/2019)	**IC** • One Laptop per Child Programme • REB e-Learning Platform • dary schools • Massive teacher training • Rwanda EdTech Hub
Senegal	SN 2025 Strategy, SIMEN, New Technology Deal	**EMIS: SIMEN (reference in Francophone Africa)** • SENPROF Digitalization, Tele-School, UNCHK • Digital classrooms & STEM labs • Priority education in SN2025
South Africa	White Paper on e-Education (2004+)	• Objective: every learner ICT-capable • • Investments in school connectivity • Provincial programs (e.g., Gauteng e-School)

TABLE OF NATIONAL REFORMS FOR THE INTEGRATION OF DIGITAL TECHNOLOGY IN EDUCATION IN AFRICA

Country	Reform / Policy	Main areas and key actions
Ghana	ICT in Education Policy (2015)	• ICT taught from primary school • iCampus GH Platform • Smart Schools • Computer skills required
Nigeria	National Digital Economy/National ICT in Education Policy	• National digital education reform • Adoption EdTech (Fellowship Mastercard • Foundation 2024) • Smart classrooms in several states
Ethiopia	ICT in Education Master Plan (2017–2025)	• E-learning universities • Digital teacher training • MoE e-learning portal • Rural inclusion
Tanzania	ICT for Basic Education (2016)	• Digital skills in the curriculum • Digital Resources TIE
Uganda	National ICT Policy (2014 – education component)	• Digitalization of the curriculum • E-learning TVET
Ivory Coast	Sectoral Education Policy (PSEF)	• ICT in secondary education • Digital Resource Centers
Mali	Educational Digital Integration Program (PINE)	• E-learning during COVID-19 • Malischool Platform
Libya	Digital Education Strategy (2020–2025)	• Post-conflict modernization • Deployment of online platforms

These reforms create a favorable policy framework for the integration of EdTech solutions into curricula and educational governance.

3. A structuring, results-oriented continental digital strategy

At the continental level, the African Union has embarked on an ambitious approach aimed at making digital technology a central driver of education system transformation. The Continental Strategy for Digital Education and Skills Development (2023–2028) now constitutes the main framework for the coordinated integration of technology into education in Africa (African Union, 2023).

This strategy aims to simultaneously address the challenges of access, quality, equity and governance, through clearly defined operational objectives:

1. Improving educational connectivity: The African Union has set a target of providing secure and reliable connectivity to at least 50% of African educational institutions by 2027, with priority given to public, rural and peri-urban schools (African Union, 20231).

2. Equitable access to digital equipment: The strategy envisions progressive access to digital devices for 20% of learners and 50% of teachers by 2027, with a ramp-up to 100% of teachers by 2030, to support blended learning and educational continuity (African Union, 2023[1]; UNESCO & AU, 2022[2]). Strengthening digital education infrastructure: The deployment of regional educational cloud infrastructure, shared platforms, and pooled digital services is identified as a key lever for improving the resilience of education systems and reducing operating costs (African Union, 2020; World Bank, 2021–2023[3]). Data governance and interoperability: The strategy emphasizes the establishment of common frameworks for educational data governance, personal data protection, and interoperability standards, which are essential for integrating EdTech solutions into existing national systems. (EMIS, LMS, SIS) (African Union, 2023[4]).

3. Digital skills development: beyond infrastructure, the strategy underlines the need to strengthen the capacities of teachers, administrative staff and public decision-makers in order to ensure sustainable appropriation of digital tools and an effective transformation of teaching practices (UNESCO & AU, 2022[4]).

Taken together, these strategic orientations lay the foundations for a more coherent, interconnected, and innovation-friendly continental educational environment, capable of supporting the emergence of sustainable, contextualized, and high-impact EdTech solutions, while facilitating the harmonization of digital education policies across the African continent.

E. CONCLUSION

A MARKET IN TRANSITION, WITH STRONG POTENTIAL

Africa is currently at a pivotal moment in its educational development. Driven by a young population, strong entrepreneurial creativity, and the rise of technological solutions tailored to local realities, the continent has an unprecedented opportunity to sustainably transform its learning systems.

[1] Union Africaine. Continental Education Strategy for Africa (CESA) – Digital Education and Skills Development Framework 2023–2028. African Union Commission, 2023.
[2] Union Africaine. Digital Transformation Strategy for Africa (2020–2030). African Union Commission, 2020.
[3] World Bank. Digital Economy for Africa Initiative (DE4A): Education and Skills Pillar. World Bank Group, 2021–2023.
[4] UNESCO & Union Africaine. Digital Education in Africa: Frameworks, Policies and Implementation Guidelines. UNESCO, 2022.



The African EdTech ecosystem, still fragmented and undercapitalized, is gradually entering a phase of structuring: the rise of innovation hubs, the proliferation of hybrid solutions, improvements in connectivity, curricular reforms integrating digital technology, and increasing commitment from international donors. All of these dynamics create a favorable context for the emergence of more inclusive, flexible educational models better aligned with the needs of a growing digital economy.

In this trajectory, the Dakar EdTech Hub, initiated by timbuktoo and supported by a pan-African network of expertise, occupies a strategic position. Its mission is clear: to catalyze the transformation of the EdTech sector in Francophone Africa. This involves:

- *the structuring of a market that is currently fragmented,*
- *the interconnection of actors (startups, states, researchers, investors),*
- *the creation of appropriate financing mechanisms,*
- *and the production of data and evidence of impact to guide public policy.*

By uniting efforts, supporting local innovations, and strengthening the capacities of educational institutions, the Dakar EdTech Hub has the potential to become a continental driver of transformation. It can contribute to building a robust, inclusive, and sustainable EdTech ecosystem capable of providing millions of young Africans with equitable access to quality learning opportunities.

Africa thus has not only strong potential, but also a historic window of opportunity to shape the future of education across the continent – a more agile, more accessible, more relevant, and truly future-oriented education.



K-12	Higher education	Professional training

GENERAL INTRODUCTION

The African EdTech ecosystem is developing according to distinct pedagogical approaches, directly linked to national priorities, learner demographics, and the varying capacities of education systems. Each segment addresses specific needs in terms of content, infrastructure, learning methodology, and business models.

The typology presented below is based on the frameworks proposed by HolonIQ, EdTech Hub, Injini, as well as sector analyses from the WAEMU and the Mastercard Foundation. It identifies three main categories that structure the EdTech offering on the continent:







A. Basic Education (K-12)

The K-12 segment: a central pillar of the African EdTech ecosystem

The K-12 segment, which covers primary and secondary education, constitutes the structural foundation of the African EdTech ecosystem. It concentrates the majority of the continent's learning population and crystallizes the main systemic challenges of African education systems: high demographic pressure, territorial inequalities, a lack of school infrastructure and a shortage of qualified teachers.

According to data compiled by the UNESCO Institute for Statistics (UIS), sub-Saharan Africa has more than 300 million children aged 5 to 14, making K-12 the largest educational segment on the continent in terms of learner numbers. This demographic trend is set to intensify: by 2050, nearly 40% of the world's school-aged children will live in Africa, increasing the pressure on national education systems.

Faced with these constraints, EdTech solutions are positioning themselves as a strategic lever for K-12 transformation, by providing:

- tools for large-scale distribution of educational content (mobile, offline, TV, radio, digital platforms);
- solutions to support teachers, including through structured content, school LMS and learning monitoring tools;
- mechanisms to reduce inequalities, by facilitating access to education in rural, isolated or fragile areas;
- more personalized and adaptive teaching approaches, sometimes enhanced by AI and educational data analysis.

In the African EdTech ecosystem, the K-12 thus appears as:

- the most mature segment in terms of use cases,
- the first point of entry for EdTech public policies,
- and the main driver of large-scale adoption, particularly through public-private partnerships.

1 In sub-Saharan Africa, the population of children aged 5 to 14 already exceeds 300 million, making K-12 the largest educational segment on the continent.
UNESCO Institute for Statistics (UIS), cited in your document.

2. **DIGITAL CONTENT ALIGNED WITH NATIONAL CURRICULA**

These solutions offer resources designed or adapted to align with official educational programs:

- *educational videos structured by subject;*
- *interactive lessons that promote engagement;*
- *formative assessment exercises and automated corrections;*
- *resources for teachers (lesson plans, guides, printable materials).*

Objective: to offer quality materials adapted to school levels, reduce inequalities in access to resources and facilitate lesson preparation.

3. **LEARNING AND REVISION PLATFORMS**

These tools enable continuous, individualized, and measurable learning:

- *Adaptive quizzes that adjust to the student's actual level;*
- *Dashboards, allowing for precise monitoring of progress;*
- *Personalized learning paths via performance analysis;*
- *Notifications and reminders to structure study time.*

Objective: to strengthen essential skills, facilitate remediation and offer differentiated instruction in often overcrowded classrooms.

4. **LOW-DATA AND OFFLINE-FIRST MOBILE APPLICATIONS**

Developed to operate in low-resource environments, they are:

- *optimized for low connectivity (compressed content, low data consumption);*
- *accessible offline via local download;*
- *compatible with entry-level smartphones, which are the majority in African households;*
- *often designed with a mobile-first approach.*

Objective: to ensure the inclusion of learners in rural areas, peri-urban areas and in households with low purchasing power.



5. **LITERACY SOLUTIONS IN LOCAL LANGUAGES**

They target the improvement of fundamental skills, particularly in literacy:

- *phonetic applications adapted to native languages;*
- *interactive educational games;*
- *culturally contextualized content;*
- *early childhood reading initiation programs.*

Objective: to improve basic learning in a context where more than 60% of 10-year-olds cannot read and understand a simple text (World Bank Learning Poverty indicator).

6. **SCHOOL MANAGEMENT SYSTEMS**

These tools digitize the administrative and management functions of institutions:

- *management of registrations, grades, attendance, exams;*
- *school-parent communication via SMS or notifications;*
- *financial and logistical management;*
- *synchronization with national systems (e.g. SIMEN, SIGE, NEMIS).*

Objective: to improve school governance, ensure data reliability, and reduce the administrative burden on teachers

7. **ADAPTATION TO THE AFRICAN CONTEXT**

All these solutions incorporate the structural constraints of the continent by design:

- *Limited connectivity: Priority given to low-data, offline, SMS/USSD and compressed resource formats.*
- *Low purchasing power: Freemium models, data-education packages, telecom partnerships, school-community pooling.*
- *Shortage of trained teachers: Turnkey formats, integrated tutorials, lesson preparation tools, digital support.*
- *Linguistic and cultural diversity; contextualized content, integration of local languages and pedagogical adaptation.*

Objective: to ensure the inclusion of learners in rural areas, peri-urban areas and in households with low purchasing power.





B. Higher Education

The African higher education sector is rapidly transforming thanks to educational technologies. Several types of solutions are structuring this transition.

1. LEARNING MANAGEMENT SYSTEMS (LMS)

Learning Management Systems (LMS) are central to the digitalization of higher education. They allow for the hosting of courses, the organization of virtual classrooms, the tracking of student progress, and the facilitation of pedagogical interactions. Common tools in Africa include Moodle, Canvas, Brightspace/D2L, as well as local LMS platforms (UNCHK, Xarala, Neolean).

2. ACADEMIC INFORMATION SYSTEMS (SIS)

Information systems (IS) modernize university administration: online registration, grade and exam management, digital diplomas, attendance tracking, and administrative dashboards. They contribute to improving academic governance and reducing administrative burdens.

3. MOOCS AND VIRTUAL UNIVERSITIES

The continent is experiencing a surge in MOOCs (MOOC Africa, FUN-MOOC, Coursera, edX) and virtual universities (UNCHK, AVU, Open University of Tanzania). These initiatives improve access to higher education and offer flexible training programs tailored to students, workers, and women.

4. REMOTE LABORATORIES AND SIMULATIONS

Digital simulation tools enable virtual practical work in science, engineering, health or agriculture.
They compensate for the lack of physical laboratory infrastructure in many African countries.

5. MICRO-CERTIFICATIONS AND DIGITAL CREDENTIALS

Micro-credentials (badges, nanodegrees, short courses) are growing rapidly in digital fields (AI, cloud computing, cybersecurity) and transferable skills. They allow for fast and targeted learning paths, aligned with the immediate needs of the job market.

Conclusion – Specific challenges of the K-12 segment in Africa

The K-12 segment represents both the greatest strength and the main area of vulnerability in African education systems. Its demographic weight, its structuring role in educational pathways, and its integration into public policies make it a strategic priority for EdTech. However, it continues to face significant systemic constraints that determine the nature, scale, and sustainability of the solutions deployed.

Current challenges

The development of K-12 EdTech solutions in Africa faces several major challenges:

- Overcrowded classrooms, inadequate infrastructure and a chronic shortage of teachers limit the quality of learning, particularly in rural and peri-urban areas.
- The gap between connected urban areas and rural or isolated areas remains marked, reinforcing the risks of educational exclusion.
- The adoption of EdTech tools by public schools often remains fragmented, dependent on pilot projects or external funding.
- The majority of K-12 solutions operate on low margins, with a strong dependence on public partnerships, landlords or telecoms.

Structuring opportunities

Despite these constraints, the K-12 segment offers unique opportunities for large-scale transformation:

- **Volume and systemic impact effect**

Any marginal improvement in access or quality has a massive impact given the number of learners.

- **Strong alignment with public priorities**

K-12 is at the heart of national education strategies, which facilitates public-private partnerships (PPPs) and nationwide deployments.

- **Accelerating post-COVID digital adoption**

The health crisis has permanently legitimized the use of digital tools in basic education.

- **Potential for hybrid solutions (tech + field)**

Models combining digital platforms, educational content and human support show better resilience.

Adaptation to the African context: a key success factor

The most relevant K-12 EdTech solutions on the continent share an “Africa-first” design, integrating local structural constraints from the outset:

- **Limited connectivity**

Priority given to low-data, offline, SMS/USSD formats, compressed content and multi-channel distribution (mobile, TV, radio).

- **Low household purchasing power**

Freemium models, low-cost subscriptions, data-education packages, partnerships with telecom operators and school-community pooling.

- **Shortage of trained teachers**

Turnkey solutions including teaching guides, integrated tutorials, structured lesson plans and tools to aid in preparation and evaluation.

- **Linguistic and cultural diversity**

Contextualized content, integration of local languages, adaptation to national curricula and socio-cultural realities.



Illustrative examples of African K-12 EdTech startups

SEVERAL AFRICAN STARTUPS EMBODY THESE DYNAMICS OF ADAPTATION AND IMPACT:

Startup	Country	Type of solution	Adaptation to the African context	Key added value
uLesson	Nigeria	K-12 platform (content + exercises)	Mobile access optimized for low bandwidth, partial offline use	Improved learning quality on a large scale through structured and interactive content
Eneza Education	Kenya	Educational content via SMS/USSD	Works without a smartphone or internet connection, low access cost	Educational inclusion of rural areas and populations with low connectivity
Kajou	Senegal	Offline content distribution	SD cards, local servers, no connectivity required	Access to K-12 content in the most remote areas
Yello App	Senegal	Mobile app for academic support	Aligned with the national curriculum, accessible design	Contextualized academic support and easy adoption by families
Nafham	Egypt	K-12 educational content platform	Arabic-language content, cultural contextualization	High adoption rate thanks to linguistic and pedagogical suitability





C. Vocational training & digital skills

The vocational training and digital skills development segment is now among the most dynamic and structuring in the African EdTech ecosystem. It is developing in response to a dual systemic pressure:

- on the one hand, the growing demand from companies for technical skills that can be directly mobilized in the digital economy (software development, data, cloud, cybersecurity, AI);

- On the other hand, the inadequacy of traditional systems (TVET, universities, public training centers) to produce, on a large scale, profiles that are immediately operational and aligned with the needs of the labor market.

In this context, EdTech players are positioning themselves as agile intermediaries between training, certification and employment, with models strongly oriented towards economic impact and the employability of young people.

1. **TYPES OF SOLUTIONS OFFERED**

The segment is structured around several categories of complementary solutions:

Category	Description	Key added value
Intensive boot camps	Accelerated programs focused on immediately employable technical skills (web and mobile development, data science, AI, cybersecurity, cloud, DevOps).	Rapid skills development, high short-term employability
Professional retraining platforms	Modular courses enabling young graduates, informal workers, or employees to retrain for careers in digital technology.	Professional inclusion and social mobility
Low-cost micro-certifications	Badges, nanodegrees, and short certificates, often aligned with market needs and recognized by companies.	Flexibility, affordability, progressive recognition
VR/AR simulations for technical training (TVET)	Simulators for industrial, agricultural, logistics, or medical professions, overcoming limitations in access to physical equipment.	Reduced equipment costs, learning by doing
Job-skill matching solutions	Platforms connecting trained talent and companies, incorporating technical tests, portfolios, or recommendation algorithms.	Accelerated professional integration

2. **SPECIFIC ISSUES**

Current challenges

Despite its dynamism, the vocational training and digital skills segment continues to face several structural constraints:

- **Limited alignment with labor market needs:** in many cases, training programs do not fully match the skills sought by companies, particularly in rapidly evolving technology sectors.

- **Insufficient institutional recognition of private certifications:** the absence of harmonized accreditation frameworks limits the official value of micro-certifications and reduces their portability across national qualification systems.

- **Unequal financial accessibility**: despite the emergence of scholarship models and inclusive pricing, the cost of some training courses remains prohibitive for some target audiences, particularly young people from disadvantaged backgrounds.

- **Lack of technical infrastructure for TVET devices:** vocational training centers often have insufficient equipment, which is a major obstacle for sectors requiring specialized laboratories, workshops or technical environments.

Opportunities

However, several favorable dynamics pave the way for a strategic strengthening of the sector:

- **Growing demand for specialized technical skills:** African companies are expressing an increased need for qualified profiles, developers, data analysts, network technicians, cloud or AI experts offering considerable potential for professional integration.

- **Persistent limitations of traditional education:** the inability of traditional education systems to absorb demand and offer suitable training creates an environment conducive to the emergence of flexible, innovative and professional EdTech solutions.

- **Political prioritization of youth employability:** at the continental and regional level (AU, WAEMU, ECOWAS), the rise in importance of policies focused on skills and professional integration constitutes an opportunity to further embed EdTech initiatives in national strategies.

- **Expansion of public-private partnerships:** the growing interest of companies, donors and governments in structuring collaborations facilitates the progressive integration of digital certifications into TVETs, universities and national qualification schemes.

3. **THE SURVEY RESULTS**

The results presented in this section are based on a pan-African survey conducted among stakeholders in the EdTech ecosystem, aimed at assessing the sector's maturity level, the educational segments addressed, the dominant business models, and the main challenges faced by EdTech initiatives on the continent. The responses collected offer both a geographical and structural perspective on the dynamics at play, particularly in Francophone Africa.

The analysis highlights a high concentration of respondents in West Africa, which constitutes the main area of participation in the survey. Three countries stand out clearly and together account for almost 60% of all responses collected.

Benin comes out on top with 27.8% of respondents, confirming the emergence of a particularly active EdTech ecosystem, supported by public policies favorable to educational innovation, the presence of structuring support structures and a strong mobilization of entrepreneurs in the sector.

Ivory Coast represents 16.7% of the responses, reflecting the gradual rise of EdTech solutions in a context marked by increasing investments in digital education and vocational training.

Senegal, with 13.9% of respondents, stands out as a strategic hub, benefiting from a structured institutional framework around the Digital Senegal Strategy, SIMEN and regional initiatives such as the timbuktoo program and the prospect of the Dakar EdTech Hub.

Beyond these three countries, responses also come from actors based in several other West, Central, East, and North African states, offering a diversity of national contexts and ecosystem maturity levels. This geographical distribution allows for the identification of common trends while highlighting significant disparities related to infrastructure availability, institutional capacity, and the regulatory environment specific to each country.



The survey results show that, in the most represented countries, EdTech players are primarily focused on the K-12, vocational training, and digital skills segments, with a strong emphasis on low-cost solutions adapted to contexts of limited connectivity. The challenges identified—insufficient funding, difficulties accessing public procurement, lack of institutional recognition, and limited human resources—are recurring issues across all geographic areas, although they are more pronounced in countries with fewer responses.

This geographical concentration of responses is not an analytical limitation, but rather a **strategic indicator:** it reflects the territories where EdTech dynamics are most visible, best structured, and most inclined to participate in regional collaborative initiatives. As such, the lessons learned from the survey provide a solid operational basis for guiding the priorities of the Dakar EdTech Hub, primarily targeting countries and ecosystems that possess a critical mass of stakeholders, the capacity to absorb innovations, and the potential for regional scaling.

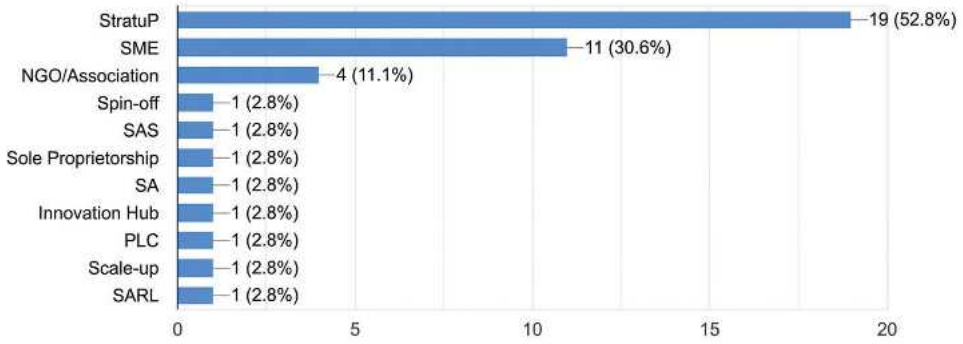


4. THE STRUCTURE OF THE SURVEY FORM.

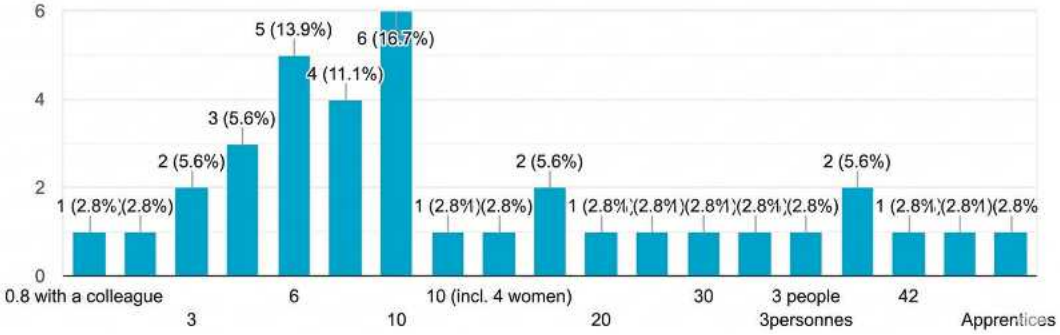
Each section presents the analysis of the data collected for:

- The general profile of the respondents and their context of creation.
- The EdTech offering includes segments (HolonIQ) and intervention sectors.
- Business models, traction and operational results.
- An in-depth synthesis of the challenges and priority needs of EdTechs for their development, as well as their recommendations for the UNDP.

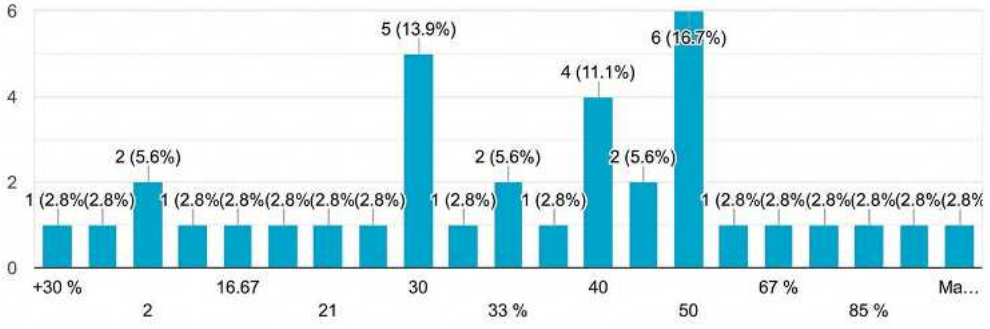
Legal Form

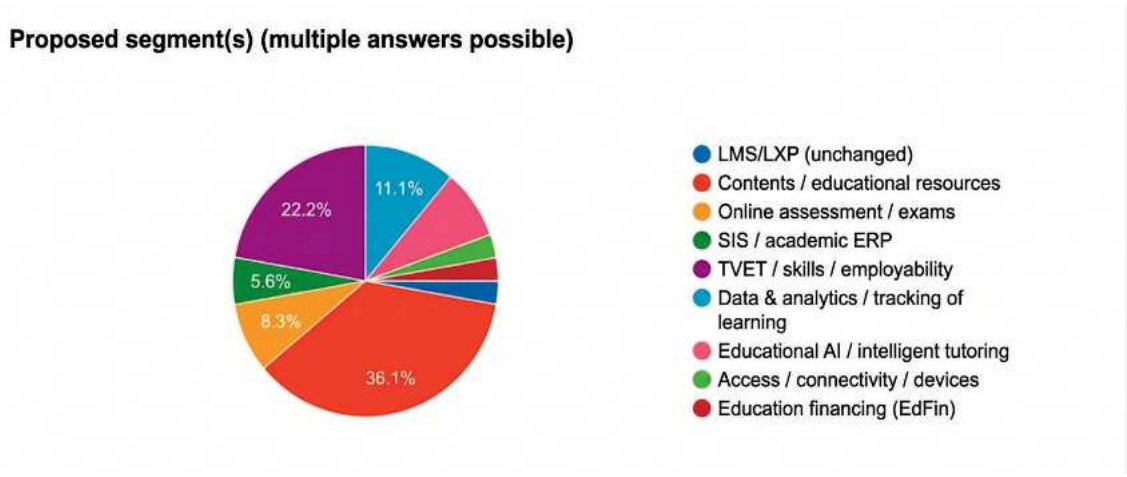


Team Size (FTE)



% of women on the team (estimate)





Profile and dynamics of the EdTech ecosystem analyzed

The EdTech ecosystem studied is primarily composed of startups (53%) and SMEs (31%), reflecting a sector that is still young, innovative, but in a phase of structuring. The presence of NGOs (11%) also underscores a strong focus on social impact and educational inclusion.

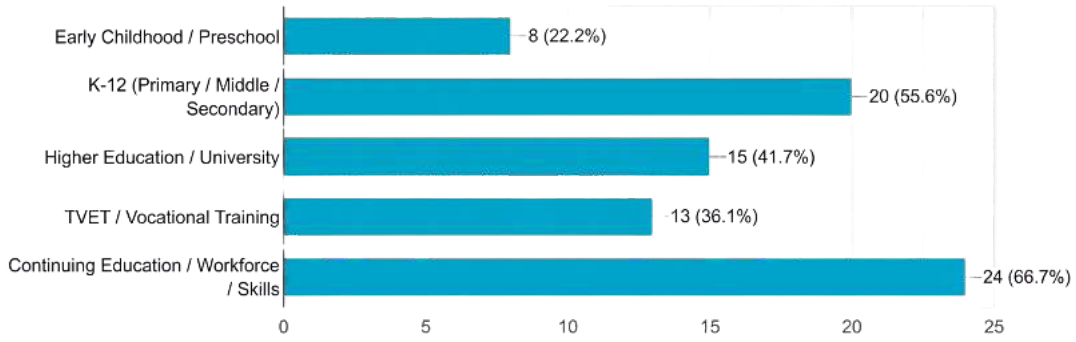
Teams are small, primarily consisting of 3 to 10 people, reflecting limited operational capacity and a strong reliance on the founders' core skills. More mature structures with larger teams remain marginal.

The representation of women is generally heterogeneous but encouraging, with many organizations having between 30% and 50% women, and a few cases of high feminization (>60%). This positions EdTech as a sector conducive to inclusion and gender equality policies.

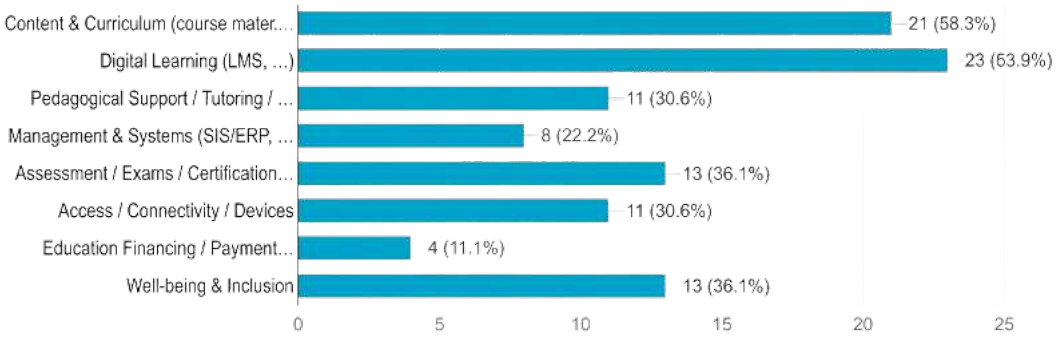
In terms of offerings, the ecosystem is heavily geared towards educational content (36%) and TVET/skills/employability (22%), segments that are quicker to deploy and less capital-intensive. Conversely, more structural solutions (advanced LMS, SIS/EMIS, educational data & AI) remain underrepresented, revealing a lack of digital educational infrastructure.

In summary, African EdTech is characterized by strong entrepreneurial dynamism and high impact potential, but suffers from a lack of institutional and technological maturity. Major opportunities lie in supporting scaling up, strengthening teams, and targeted investment in key sectors (data, education systems, AI).

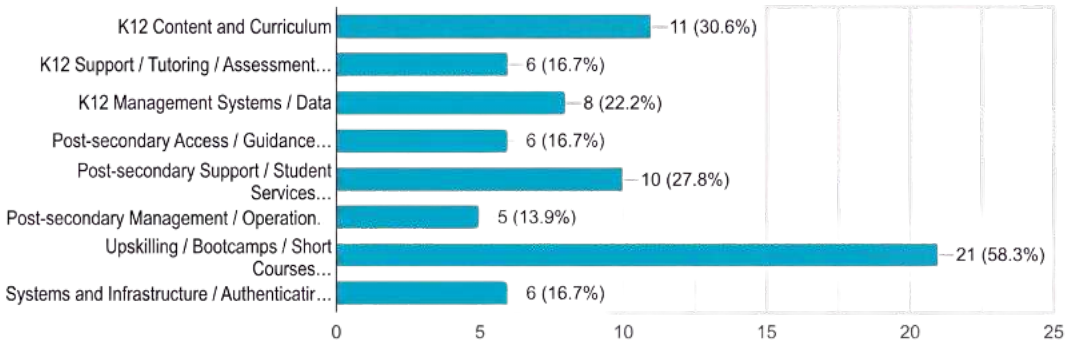
Principal Sectors (1 choice)



Sub-sector / Primary function (1 choice)



Nearest HolonIQ Cluster (1 choice)



Main sector.

The ecosystem is dominated by continuing education/workforce/skills (66.7%), followed by K-12 (55.6%) and higher education (41.7%). TVET is well represented (36.1%), while early childhood education remains a minority (22.2%).

- **Key takeaway: strong focus on employability and retraining, with a K-12 foundation still central.**

Sub-sector / main function.

Priorities are focused on digital learning (LMS/LXP – 63.9%) and content & curriculum (58.3%). These are followed by assessment/certification and well-being & inclusion (36.1% each). Management systems (SIS/ERP – 22.2%) and education financing (11.1%) are underrepresented.

- **Key reading: pedagogy and delivery-first approach, lack of infrastructure and financial solutions.**

Nearest HolonIQ cluster.

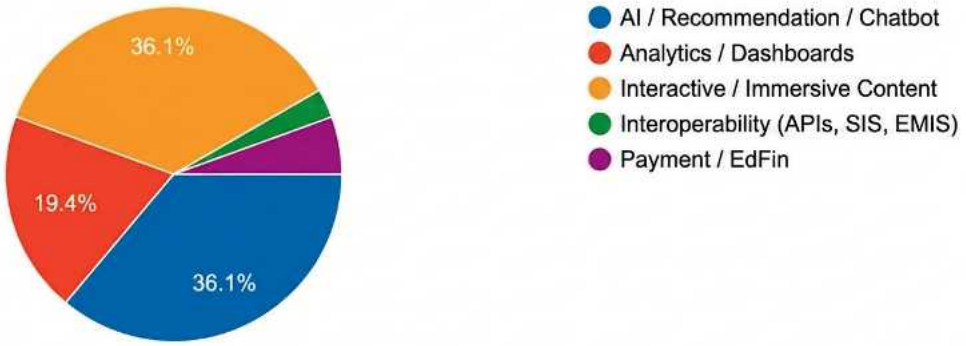
The dominant cluster is Upskilling / Bootcamps / Short courses (58.3%). For K-12s, Content & Curriculum (30.6%) and Management/Data (22.2%) stand out. In post-secondary education, student support (27.8%) is more popular than management systems (13.9%).

- **Key reading: clear specialization in the short term, operational and employability, with potential for development on structuring systems.**

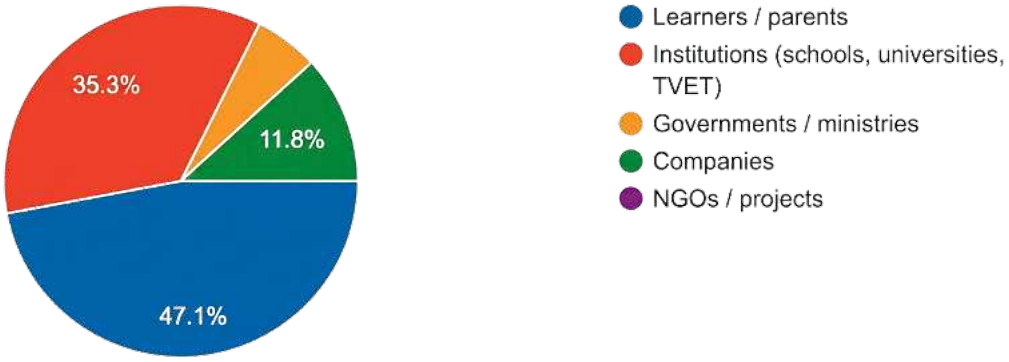
Cross-sectional conclusion.

An ecosystem geared towards rapid impact and employability, rich in content and learning platforms, but with blind spots in systems (SIS/EMIS), EdFin and infrastructure—key opportunities for the next phase of structuring

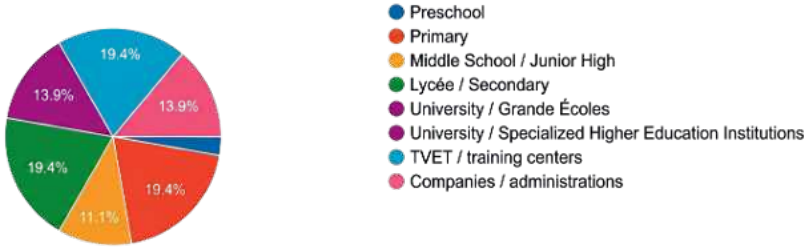
Innovations / Key Technologies (multi)



Target audiences (multi)



Level(s) of education actually served (multi)



Key innovations/technologies.

The solutions primarily focus on AI (recommendations, chatbots) and interactive/immersive content (36.1% each), reflecting a strong emphasis on the learning experience and personalization. Analytics and dashboards occupy an intermediate position (19.4%), while system interoperability (APIs, SIS, EMIS) and payment/EdFin remain marginally addressed.

- **Key takeaway: innovation focused on educational use, low maturity on systemic and financial building blocks.**

Target audiences.

Learners and parents are the primary target audience (47.1%), followed by educational institutions (35.3%). Businesses (11.8%) and governments/ministries remain secondary.

- **Key takeaway: predominance of B2C and B2B-education models, with low penetration of B2G.**

Education levels served.

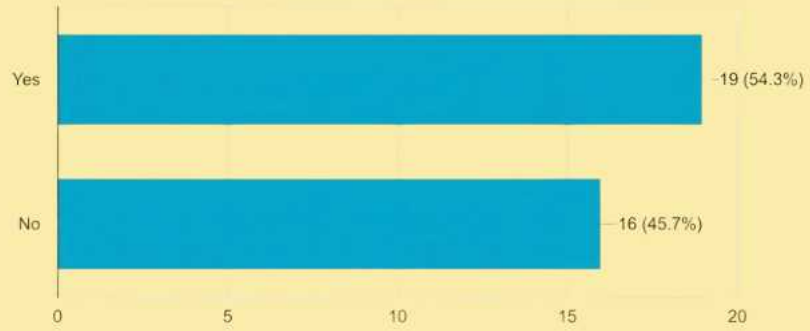
Coverage is relatively balanced between primary, secondary, and TVET/training centers (approximately 19% each). Colleges are less covered (11.1%), while universities/higher education institutions and businesses/government agencies each represent 13.9%. Preschool remains marginal.

- **Key takeaway: focus on high-volume levels and immediate impact, with potential for development into preschool and institutional solutions.**

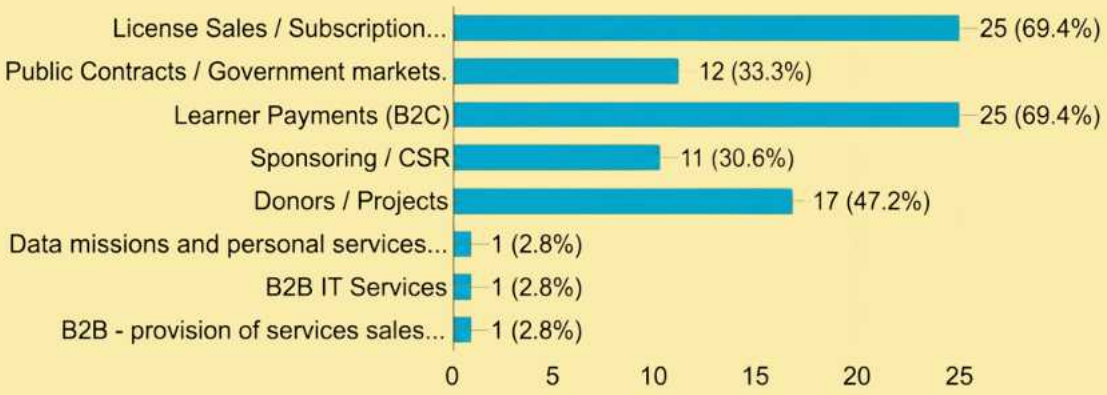
Cross-sectional conclusion.

The ecosystem favours learner-oriented educational innovations, primarily addressing the direct market (learners/schools), but still leaves infrastructure technologies, interoperability and EdFin *(the main levers for systemic scaling)* behind.

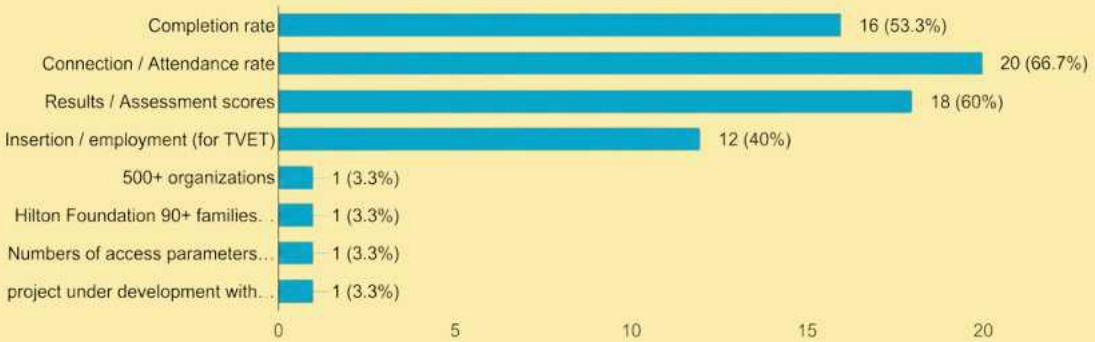
Have you benefited from financing / investment?



How do you generate revenue?



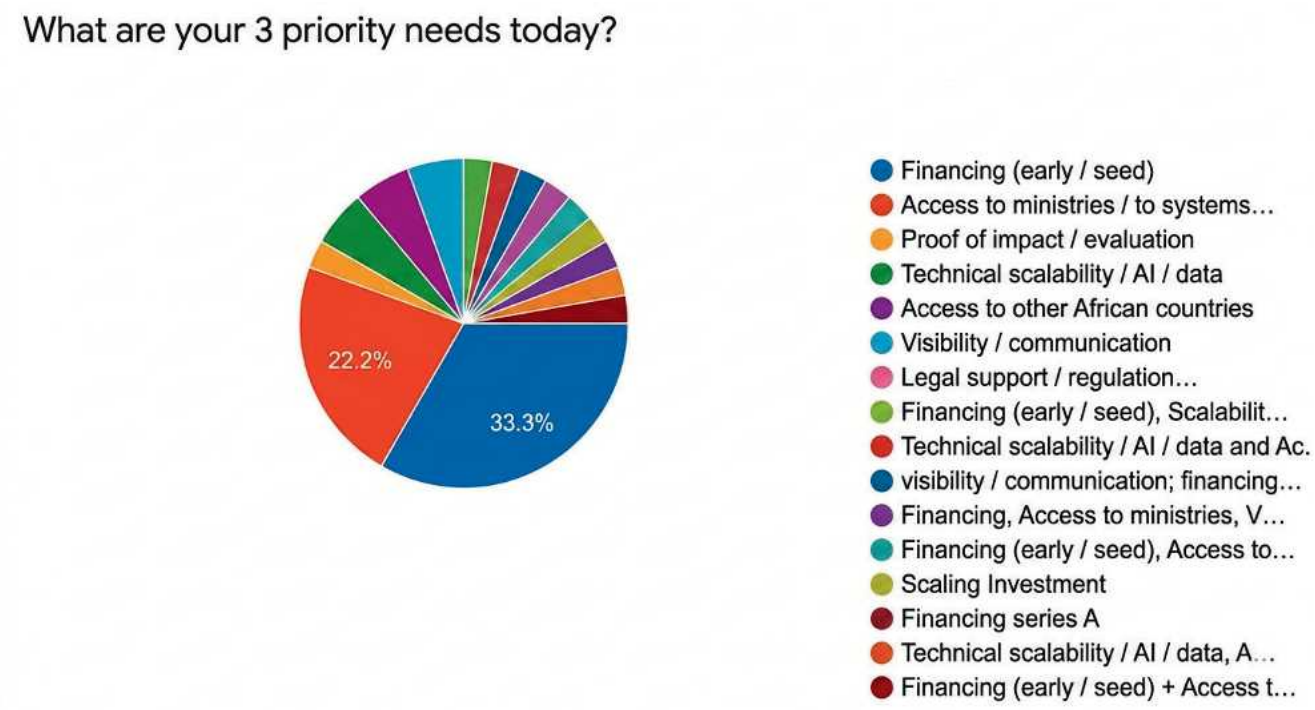
. Number of clients/organizations served



Brief interpretation

- **Funding:** The ecosystem is partially funded (54%), but nearly one in two players remains without access to capital, revealing a gap in seed and scale.
- **Business models:** Revenue is primarily based on B2C (learner payments) and subscriptions/licenses, with significant continued reliance on donors and projects. B2B/B2G models remain underutilized.
- **Traction and performance:** Stakeholders primarily measure engagement and educational performance (connection, results, completion). The employment impact, key for TVET, is still insufficiently monitored.
- **Regulatory constraints:** The majority have not encountered major obstacles, but a third face constraints related to regulation, data and public procurement, hindering systemic solutions.

Overall reading: A dynamic and user-oriented EdTech ecosystem, but still financially fragile, with a clear need for structural funding, revenue diversification and better regulatory alignment for scaling up.



Availability for data sharing.

The support is almost unanimous: 94.4% of respondents are ready to share anonymized data for pan-African mapping.

- **Key takeaway:** high level of trust and collective maturity; the conditions are in place to build a credible continental database to support the structuring of the EdTech ecosystem.

Priority needs expressed.

1. Funding (early / seed) – 33.3%: main obstacle to growth and sustainability.
2. Access to ministries and public education systems – 22.2%: difficulty integrating into public policies and markets.

Technical scalability / AI / data (and evidence of impact, visibility): cross-cutting needs to scale up regionally.

- **Key takeaway:** actors are not just asking for capital, but for systemic access (states, data, regulatory frameworks) and advanced technical support.

Strategic conclusion.

The African EdTech ecosystem is open to data sharing and expresses clear needs for funding, institutional integration, and technological scaling. This argues for coordinated continental mechanisms (funds, hubs, data frameworks, public partnerships) rather than isolated interventions.



3. MAPPING BY GEOGRAPHIC REGION: EDTECH GROWTH HUBS IN AFRICA

GENERAL INTRODUCTION – ONE CONTINENT, DIFFERENT EDTECH TRAJECTORIES

The development of EdTech in Africa is not following a homogeneous trajectory. While it is part of an overall growth dynamic driven by demographics, the widespread adoption of mobile devices, and the urgent need to transform education systems, it nevertheless reveals significant regional and national disparities.

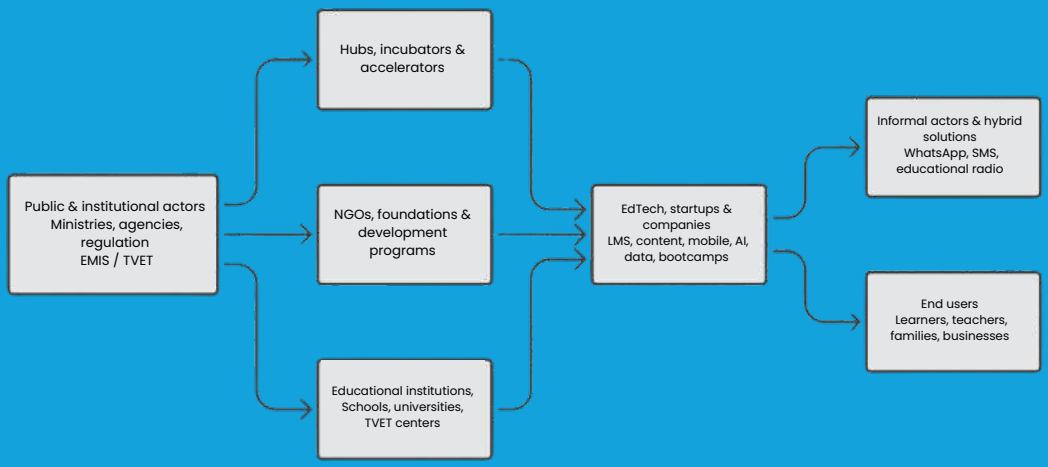
These differences can be explained by a combination of structural factors: level of digital maturity, quality and coverage of infrastructure (connectivity, energy, equipment), existence of regulatory frameworks favorable to educational innovation, vitality of technological ecosystems, as well as the intensity of public, private and philanthropic investments.

It is essential to emphasize that the African, and particularly West African, EdTech ecosystem is not limited to startups. It is based on a range of diverse and interconnected actors, including:

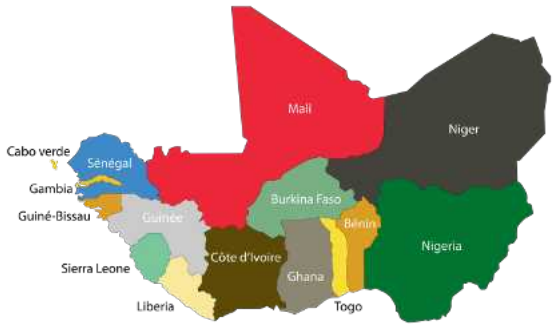
- EdTech startups and companies,
- public institutions (ministries, agencies, regulatory bodies),
- educational institutions (schools, universities, TVET centers),
- NGOs, foundations and donor-funded projects,
- innovation hubs, incubators and accelerators,
- as well as informal or hybrid actors using digital tools for educational purposes (mobile learning, WhatsApp, SMS, educational radio, community platforms).

This diversity of actors shapes specific EdTech trajectories, where innovation is manifested not only by the creation of technology startups, but also by the progressive adaptation of existing educational systems, the experimentation of hybrid public-private models, and the integration of large-scale digital tools in contexts of constrained resources.

In this context, EdTech appears less as a single market than as a set of regional sub-ecosystems, strongly influenced by public policies, demographic dynamics, institutional capacities to absorb innovation and the structuring role of technical and financial partners.



A. West Africa



A demographic pool with strong potential, but an EdTech ecosystem that is still fragmented.

1. General characteristics

West Africa represents one of the main areas of potential growth for EdTech in Africa, driven by exceptionally dynamic demographics and rapidly expanding educational demand. The WAEMU region, in particular, has a very large young population, with education systems facing increasing structural pressures (mass schooling, teacher shortages, limited infrastructure).

Despite this potential, the West African EdTech ecosystem remains in a structuring phase. It is characterized by an uneven adoption of educational technologies, heavily conditioned by infrastructure, financing and governance constraints.

The widespread use of mobile solutions (smartphones, SMS, WhatsApp, lightweight applications) is due to limited fixed connectivity and uneven broadband network coverage. EdTech innovation in these areas is therefore largely frugal, mobile-first, and context-dependent, but rarely standardized or interoperable on a regional scale.

Major structural challenges

- **School connectivity is still low, often less than 20% in several countries (particularly in rural areas).**
- **Access to digital technology is primarily via mobile, with limited data usage.**
- **Strong dependence on funding from international donors, particularly for pilot projects and large-scale deployments.**
- **The EdTech offering is fragmented, poorly coordinated, weakly standardized, and still difficult to scale regionally.**

2. **KEY REGIONAL TRENDS**

Several cross-cutting trends are emerging across the region:

- ***Massive use of mobile-first solutions, adapted to connectivity constraints.***
- ***Rapid development of contextualized content, increasingly including local languages.***

- **Explosion of bootcamps and short training courses, in response to the challenges of youth employability.**
- **Increase in informal educational uses, particularly via WhatsApp, SMS, audio and community micro-learning.**

3. **SPECIFIC CONSTRAINTS IN WEST AFRICA**

Despite its dynamism, the region faces persistent structural constraints:

- Fragmentation of the Francophone market, with regulatory frameworks and curricula that are poorly harmonized.
- Low direct public investment in educational innovation and EdTech.
- Absence or weakness of national frameworks dedicated to EdTech, limiting the systemic integration of solutions.
- Marked heterogeneity in connectivity and institutional capacities between countries.

Conclusion: Strategic reading

West Africa boasts a dynamic, innovative, and initiative-rich EdTech ecosystem, but it remains insufficiently structured and coordinated at the regional level. The potential for scaling up is real, but contingent upon better harmonization of frameworks, increased public investment, and greater coordination among stakeholders.

➡ In this context, the Dakar EdTech Hub appears as a strategic regional infrastructure, capable of:

- to unite the diversity of actors (public, private, hubs, NGOs),
- harmonize approaches and standards,
- to facilitate public-private dialogue,
- and accelerate the scalability of West African EdTech solutions

B. East Africa

The Reference Centre for Technological Maturity in Education.

1. General Characteristics and Authorization Framework

The East African region has established itself as the continent's leading EdTech innovation hub, often referred to as the birthplace of this dynamic. This emergence is based on a convergence of highly favorable structural factors, including a political environment open to reform, an advanced digital culture among the population, particularly dense technological ecosystems, and sustained investment commitments.

REGIONAL DRIVING FORCES AND DYNAMICS:

Governance and Public Policy: Advanced integration and proactive adoption of policies related to Information and Communication Technologies for Education (ICT4E).

PARTNERSHIPS AND FUNDING:

Mobilization of significant funding from international donors and development partners, ensuring growth capital.

ECOSYSTEM CONCENTRATION:

Presence of a significant density of digital hubs (Kenya, Rwanda, Uganda) promoting the incubation and scaling of solutions.

REGULATORY FRAMEWORK:

The establishment of a favorable and proactive regulatory framework is essential for the experimentation and rapid expansion of digital platforms.

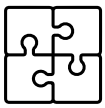


2. Analysis of Key National Actors

Country	Strategic Role	Examples of Key Players and Solutions
KENYA	Continental Leader: Driving innovation with accessibility-focused solutions.	Eneza Education (learning via SMS/USSD); BRCK / Kio Kit (solutions dedicated to rural areas); Kytabu / eLimu (digital libraries and interactive content); Ubongo Kids (educational media).
RWANDA	Institutional Reference: It is distinguished by strong and structured governmental integration.	Rwanda EdTech Hub (MCF & MINEDUC collaboration); ICT Chamber (State/innovation interface); Uji (national learning platform); Ambitious national ICT4E projects.
UGANDA	Emerging Dynamics: Focused on fundamental challenges and vocational training.	Emerging literacy solutions and expansion of digital innovations in Technical and Vocational Education and Training (TVET).

3. Strategic Directions and Trends

Regional trends confirm a trajectory of professionalization and consolidation of EdTech:



SYSTEMATIC INTEGRATION

Growth in the integration of educational technologies into national public policy frameworks.



QUALITY AND STANDARDIZATION

Strengthening efforts to standardize and certify digital educational content.



RESILIENCE OF MODELS

Massive adoption of blended learning models effectively combining online and offline modalities.



DATA-DRIVEN MANAGEMENT

Increased deployment of learning analytics to optimize learning paths, refine skills assessment and inform policy decisions.

4. **Structural Challenges and Operational Constraints**

Despite this maturity, the region faces two major obstacles that could compromise the sustainability and autonomy of the ecosystem:

Infrastructure costs: The high costs of maintaining and renewing digital infrastructure (hardware and connectivity) constitute a financial burden.

Viability and External Funding: A persistent dependence on external funding raises questions about the long-term sustainability and autonomy of EdTech initiatives.

Conclusion and Recommendation

East Africa offers a compelling model of EdTech maturity, the result of strong political will and effective collaboration between public and private actors. This regional experience should serve as a strategic benchmark for other growth hubs, particularly the Dakar Hub. The lessons learned focus specifically on the institutional integration of EdTech solutions and the structuring of robust, scalable public-private partnerships.

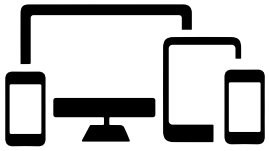


C. North Africa

A Solventful, Structured, and Rapidly Digitalizing Market: The North African region (Morocco, Tunisia, Egypt) represents a high-potential EdTech market, characterized by mature infrastructure and a demand for quality digital learning solutions. It constitutes a strategic area for investment and the development of scalable EdTech models.

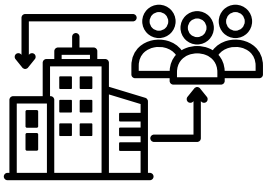
1. Structural Characteristics and General Environment

North Africa benefits from a socio-economic and technological environment that is particularly conducive to the rapid development of EdTech solutions:



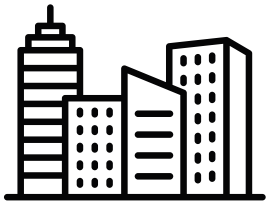
MATURATION OF DIGITAL INFRASTRUCTURES

The region boasts superior technological infrastructure, resulting in denser internet coverage and higher speeds, facilitating the deployment of data-intensive solutions.



HIGH HOUSEHOLD SOLVENCY

A relatively higher standard of living and solvency of households allows the massive adoption of B2C (Business-to-Consumer) economic models based on subscriptions and premium services (learning payments).



URBAN DYNAMISM AND CONNECTIVITY

The high rate of urbanization, combined with high internet penetration, provides fertile ground for the rapid scalability of online platforms.



PRIVATE UNIVERSITY ECOSYSTEM

The existence of a dynamic private higher education and vocational training ecosystem promotes the integration of innovative technologies and the experimentation of new educational models.

2. **Key Actors and Initiatives by Country**

The regional ecosystem is structured by powerful players focusing on content, tutoring, and vocational training:

Countries	Notable Actors & Trends
Egypt	Nafham (major educational platform); Orcas (online tutoring); Oschool (remediation); Akhdar (gamified content). The market is characterized by a large population size and a massive demand for academic support solutions.
Tunisia	GoMyCode (bootcamp leader in Africa); Schoolinka. We note the emergence of initiatives based on immersive technologies (VR/AR) in technical and vocational training (TVET).
Morocco	Strong presence of private EdTechs, rapid expansion of blended learning models and the establishment of structuring public-private partnerships for digital integration in the education sector.



3. Regional Trends and Growth Dynamics

Several market dynamics are particularly noteworthy and are shaping the region:

Accelerated growth of B2C offerings: The market is dominated by solutions directly targeting learners and parents, taking advantage of household solvency.

Generalization of the Hybrid Model (Blended Learning): The adoption of hybrid learning models, combining face-to-face and online learning, is accelerating, ensuring a gradual transition to digital while maximizing educational impact.

Content Industrialization: The proliferation of Arabic-language content and platforms is a strong trend, targeting a unified and massive linguistic market.

4. Specific Constraints and Challenges

Linguistic Fragmentation: The coexistence of Arabic, French and English, and even Amazigh, creates challenges in content localization and double investment.

Direct International Competition: As the market is more structured and solvent, it is subject to direct and fierce competition from international EdTech platforms, complicating the emergence and financing of local champions.

Conclusion

North Africa is a structured and financially sound region, offering strong opportunities for content industrialization and the adoption of consumer-oriented EdTech strategies. EdTech companies there benefit from a solid technological foundation but must strategically position themselves against international competition to ensure their scalability. This region represents a high-value entry point for investments targeting the achievement of a critical mass of users.



D. Southern Africa: An Advanced Ecosystem Dominated by South Africa

The Southern African region, and more specifically South Africa, constitutes the most mature and technologically advanced EdTech ecosystem on the continent. It is distinguished by a structured market, easy access to financing and a strong culture of lifelong learning.

1. Structural Characteristics and General Environment

Technological Maturity and Infrastructure: The region is recognized for its advanced technological maturity, offering high-quality digital infrastructure that facilitates the deployment of sophisticated solutions (VR/AR, artificial intelligence).

Solid B2B Market Structure: The market structure is characterized by strong demand for B2B (Business-to-Business) solutions, particularly for higher education and corporate learning.

Easier Access to Funding: The ecosystem benefits from easier access to private funding and venture capital, enabling EdTechs to achieve significant growth milestones.

Academic and Professional Expertise: The presence of top-level universities and a well-established culture of professional certification ensure a constant demand for formal learning and skills development solutions.

2. Key Players and Innovation Infrastructures

South Africa is the hub of this region, hosting large-scale platforms and support structures:

Country/Hub	Notable Players & Trends
South Africa	Large-scale platforms like Snapplify and Xander; major educational groups like AdvTech Group; and globally recognized online training providers such as GetSmarter (2U) and Valenture Institute are all present in the region. The region is also supported by innovation catalysts like Injini EdTech Accelerator and hubs such as Silicon Cape and Workshop17.
Other Countries (Namibia, Botswana, Zambia)	We are seeing a gradual emergence of initiatives focused on integrating digital technology into technical and vocational education and training (TVET), with ongoing government digitalisation programmes.

3. **Regional Trends and Innovations**

Several market dynamics indicate the premium and technology-oriented positioning of this region:

Development of Cutting-Edge Solutions: Southern Africa is a fertile ground for experimentation and development of Virtual Reality (VR) and Augmented Reality (AR) based solutions in the fields of technical training.

Premium EdTech Offer: The solvency of the market allows the development and adoption of a high-quality EdTech offer, often with high added value.

Continuing Education Dynamics (Corporate Learning): Continuing education and skills improvement in companies (upskilling and reskilling) constitute a major growth segment.

4. **Specific Constraints and Challenges**

The region's main challenge is linked to socio-economic disparities:

Inequalities in equitable access: Despite technological maturity, strong socio-economic inequalities limit equitable access to quality EdTech resources for disadvantaged populations or rural areas.

Dependence on the Private Sector: The viability of the products is heavily dependent on the commitment and investment of the private sector, making the systemic impact on public education slower.

Conclusion

Southern Africa represents an advanced and fertile technological ecosystem for technical innovation, co-creation of premium content, and international collaborations. It is the ideal place to test advanced continuing education (upskilling) models and sophisticated technological solutions before their potential deployment in other regions of the continent.



4. COMPARATIVE SYNTHESIS OF EDTECH ECOSYSTEMS BY REGION

The following table presents a summary of the main findings of the regional analyses, offering a synthetic view of the strengths and weaknesses of the various EdTech markets in Africa.

Region	Level of Maturity	Dominant Players and Key Examples	Key Strategic Opportunities	Strategic Opportunities
West Africa	Emerging	uLesson, Yello Xarala, LOGECO, EtriLabs	High fragmentation linguistic and regulatory fragmentation; access to low-income funding.	Massive youth population; development of mobile-first solutions; growth of the English-speaking and French-speaking markets.
East Africa	Very Mature	Eneza Education, Kytabu, Zeraki, ICT Chamber	High infrastructure costs infrastructure and bandwidth bandwidth; low purchasing power in certain areas	Public policies Advanced EdTech (Kenya, Rwanda); strong adoption of mobile technology.
North Africa	Structured & Solvable	Nafham, Orcas, GoMyCode	Strong competition from international giants; linguistic (Arabic / French /English).	Rapid expansion of B2C models; industrialization of Arabic-language content; solvent market for premium solutions.
South Africa	Technologically advanced	Snapplify, Xander, Injini Accelerator	Significant socio-economic inequalities limiting equitable access; dependence on private investment.	Innovation in VR/AR and premium content; strong momentum upskilling and continuing education; international co-creation international.

SECTION 3

TECH HUBS, INCUBATORS AND ACCELERATORS: ENGINES OF INNOVATION

1. THE ROLE OF TECH HUBS IN THE AFRICAN DIGITAL ECOSYSTEM

Tech Hubs are a strategic pillar of digital innovation on the continent. In the EdTech sector, their importance is even more pronounced, as they help to bridge persistent systemic gaps: market fragmentation, limited availability of suitable funding, limited entrepreneurial capacity, and a disconnect between innovation and educational policies.

MAIN FUNCTIONS OF TECH HUBS

A. Development of entrepreneurial skills



- Strengthening skills in product design, prototyping and user testing. Adopting lean startup methodologies.
- Financial structuring, formalization of business models and preparation for fundraising. Individualized support through coaching and mentoring.

B. Market access and strategic networking

Active connection between startups, educational institutions, investors, public operators and funders. Facilitation of pilots, experiments and proofs of concept within educational institutions. Reduction of information asymmetries between startups and public administrations.

C. Experimentation and production of innovation



Establishment of EdTech Labs, living labs and dedicated user testing spaces. Validation of solutions in real-world conditions in schools and training centers. Acceleration of action research and evidence-based approaches.

D. Structuring the ecosystem

Facilitating communities of innovation and practice. Organizing events (bootcamps, hackathons, policy labs, festivals). Technology monitoring, knowledge production and advocacy with public authorities.



Strategic implication for Dakar: the future EdTech Hub timbuktoo will have to fully embody these four functions in order to become a catalyst for educational innovation for all of Francophone Africa.



2. TYPOLOGY OF SUPPORT STRUCTURES

The African ecosystem consists of a diversity of structures, classified into three main categories:

	Incubators	Accelerators	Coworking spaces
Objective	Support startups in the ideation or pre-seed phase.	Support startups that already have commercial traction.	Promote interaction, networking, and a culture of innovation.
Key features	• Long-term programs (6 to 12 months) • Development of MVPs (Minimum Viable Products) • Structuring of business models • Mentoring and methodological support	Intensive programs (6 to 12 months) • Connecting with investors, lenders, and distribution networks • Structuring KPIs, growth models, and expansion strategies • Support for internationalization	• Hosting of local ecosystems • Access to shared resources and infrastructure • Facilitation of entrepreneurial communities • Organization of events open to the public
Relevance for EdTech	EdTech projects generally take longer to mature than those in other sectors (fintech, agritech) due to educational, institutional, and regulatory constraints. Incubation is therefore essential for testing solutions, refining business models, and demonstrating educational relevance.	Enabling promising EdTech startups that have validated their value proposition to reach national or regional scale and prepare for structural financing (Series A/B).	Coworking spaces are gateways to the EdTech ecosystem, promoting the emergence of ideas, interdisciplinary collaboration, experience sharing, and the integration of project leaders into local and regional innovation networks.
Examples in Africa	Jokkolabs (Senegal) Orange Digital Center (multiple countries) Sèmè City (Benin) Impact Hub Abidjan	Injini EdTech Accelerator (South Africa) EtriLabs / EdTech Fellowship (Senegal, Benin) CcHub Education Practice (Kenya, Nigeria) MEST Africa (Ghana)	CTIC Dakar Jokkolabs WorksUp Abidjan iHub Nairobi CcHub Nigeria

3. CASE STUDIES OF EDTECH INNOVATION HUBS: STRUCTURING MODELS AND STRATEGIC LESSONS

Analysis of EdTech ecosystems across Africa reveals the emergence of several key innovation hubs. These hubs provide essential frameworks for incubation.

A. iHub (Kenya) – The Pioneer of Innovation in East Africa

The iHub in Kenya has established itself as a key player in the East African technology landscape. Its model is distinguished by an integrated approach and strong ties to public institutions. Structure and Offerings: The Hub offers a comprehensive range of services including co-working, startup incubation, applied research, and the development of public-private partnerships. Institutional Contribution: It maintains a close and formalized collaboration with the Ministry of Education, facilitating the production and adoption of high-impact solutions (e.g., BRCK, Eneza Education). Key Strategic Lesson: The importance of a strong institutional link with the government and the adoption of a hybrid model



Key Strategic Lesson: The importance of a strong institutional link with the State and the adoption of a hybrid model combining technical incubation, educational research and advocacy activities is crucial to ensuring the relevance and national adoption of innovations.

B. CcHub (Nigeria / Kenya) – The Pan-African Impact Network

The Co-creation Hub (CcHub) represents a cross-border network model capable of operating on a continental scale, benefiting from the commitment of global strategic partners.

Scope and Specialization: The CcHub is a pan-African network with specialized programs in education and social innovation, enabling the pooling of resources and knowledge.

Structuring partnerships: Its collaborations with leading organizations (Google, Meta, Mastercard Foundation) give it a unique financial and technical capacity to operate and deploy solutions across the continent.

Key Strategic Lesson: It is imperative to strengthen the capacity of EdTech startups to produce scientific evidence, rigorously evaluate and demonstrate social and educational impact, thereby ensuring the credibility and reproducibility of innovations.

4. STRATEGIC CHALLENGES AND LESSONS FROM THE HUB ECOSYSTEM FOR THE DAKAR EDTECH HUB

Analysis of the startup support ecosystem (incubators, accelerators, hubs) on the continent reveals structural challenges that must be taken into account in the design of the EdTech Dakar Hub (timbuktoo) in order to ensure its impact and sustainability.

A. Structural Challenges Identified in the Hub Ecosystem

Traditional support structures in Africa face several limitations that hinder the effectiveness of supporting EdTechs:

• Generalist structures and scarcity of specialization: The vast majority of hubs and accelerators are generalist. The scarcity of structures specializing in EdTech prevents targeted sector-specific support, particularly regarding regulatory specificities (content approval, educational data) and business models specific to the public education sector.

- ***Vulnerability of business models: The sustainability of hubs is often precarious. Their business models are largely dependent on external funding (donors, grants) which is inherently unstable, thus limiting their ability to offer long-term support and develop specialized expertise.***

• Limited interaction with the public sphere: There is a disconnect between innovation structures and public institutions. This limited interaction reduces the potential impact of these hubs on the formulation of educational policies and the facilitation of public-private partnerships.

• Lack of Evidence-Based Data: Insufficient production and collection of usage, retention, and learning impact data hinders the demonstration of the value of solutions. This makes it difficult to guide public policy decisions and attract evidence-based investment (impact evidence/evaluation, priority needs).

B. Strategic Lessons for Positioning the Dakar EdTech Hub

To effectively meet the sector's needs (Financing, Access to ministries, Scalability) (2) and align with timbuktoo's long-term investment vision, the future EdTech Hub Dakar must adopt a logic of disruption with generalist models:

Strategic Lesson	Key Action for the EdTech Hub Dakar	Objectif (Impact)
Strong Sector Specialization	To provide high-level expertise focused on education, applied technologies, and public policy.	To provide relevant support for the specificities of EdTech and to strengthen the Hub's legitimacy with the ministries.
Setting up a Policy Lab	Create a permanent space for dialogue and regulatory co-creation with the Ministries of Education and Vocational Training.	Remove administrative and regulatory barriers (approval, public procurement) and facilitate access to public systems.
Creation of an Evidence Lab	Establish a mechanism for producing data on usage, retention, and impact on learning.	Establishing proof of impact is essential for the confidence of funders and adoption by the public sector.
Comprehensive Support Offer	To offer a continuum of services covering incubation, acceleration and post-acceleration to support growth at all stages.	Addressing the need for Funding (early/seed) and the technical/data Scalability challenges, identified as priorities.
Strong Regional Roots	To position itself as a connection point for the EdTech ecosystems of the WAEMU and Francophone West Africa.	Facilitating expansion to other African countries, thus addressing a priority need.
Timbuktoo and Catalytic Capital Alignment	To sustainably support EdTech solutions by aligning with the catalytic capital dynamic (USD 350M) of the Timbukto fund	Ensuring the sustainability of financial and strategic support for EdTech champions.

SECTION 4

STRUCTURING POLICIES, PROGRAMS AND FUNDING

1. THE LACK OF STRUCTURED EDTECH PUBLIC POLICIES: A SYSTEMIC OBSTACLE

Despite a growing awareness of the importance of digital technology in education, most African countries still lack specific public policies dedicated to EdTech. This gap results in a fragmented ecosystem, where private initiatives develop without a national governance framework or public support mechanisms, thus limiting their viability and scalability.

Observed deficits:

- Lack of national budget lines for the acquisition or deployment of EdTech solutions
- Lack of national technical and educational standards
- Purchasing procedures poorly adapted to innovative solutions
- Weak coordination between ministries (Education, Digital Technology, Finance)
- Lack of reliable national data on the use and impact of educational technologies

➡ These shortcomings reinforce dependence on international partners, weaken digital sovereignty, and slow down the integration of EdTech into educational systems.

2. EXAMPLES OF ADVANCED NATIONAL POLICIES

Several African countries stand out for developing integrated EdTech strategies, often supported by multilateral partnerships:

RWANDA An ambitious and operational ICT4E strategy	**KENYA** The CBC model and the competency approach	**SENEGAL** A digital ambition still in transition
• Explicit integration of EdTech into the education sector plan • Clear objectives regarding infrastructure, content and teacher training • Creation of a national EdTechHub in partnership with the Mastercard Foundation and the government • Field feedback mechanisms and solution validation	• Gradual rollout of the Competency-Based Curriculum (CBC) • Integrating digital resources into fundamental learning • Strong public-private partnership with Safaricom, Eneza, etc. • Ongoing evaluation of the impact of EdTech through educational inspections	• Existence of the Digital Senegal 2025 Strategy • Launch of the Diamniadio Digital Hub and the UNCHK University • There is currently no unified national EdTech framework. • A strategic opportunity for the Timbuktoo Dakar Hub to catalyze a structured EdTech policy

Quantitative survey data (Question 8.1 - Priority needs; Question 7.2 - Regulatory constraints) used to justify key actions. Source document of the survey (Survey form - EdTech, UNDP Study – Strategic mapping of the EdTech ecosystem in Africa / EdTech Hub Dakar (timbuktoo)) defining the framework of the project.

3. CONTINENTAL AND MULTILATERAL STRUCTURING PROGRAMS

Several continental and international initiatives aim to create an enabling environment for EdTech development in Africa: A. timbuktoo – UNDP (United Nations Development Programme) The timbuktoo program, launched by the UNDP, is currently one of the most ambitious initiatives in Africa to catalyze technological innovation and strengthen entrepreneurial ecosystems. Designed as a pan-African network, it is based on the creation of ten thematic hubs distributed across the continent, each dedicated to a strategic sector, including the EdTech Hub based in Dakar. This structure aims to stimulate the emergence of innovative solutions adapted to African challenges, while fostering synergies between countries and regions.

timbuktoo is leveraging an unprecedented investment strategy, with a target of $350 million in patient capital to support high-potential African startups. This approach allows for funding not only seed and growth stages, but also the consolidation of technology companies in sectors deemed priorities for the continent's economic and social development.

The program's ambition goes beyond mere funding: timbuktoo seeks to create an environment conducive to innovation by combining infrastructure, strategic support, strengthened institutional capacities, and public-private partnerships. Within this framework, the Dakar EdTech Hub occupies a particularly strategic position. It is expected to play a regional coordinating role, supporting educational startups, structuring a still fragmented market, and facilitating dialogue between public, private, and academic stakeholders in the region.

Thus, timbuktoo presents itself as a catalyst for large-scale transformation, aiming to make Africa a space where technological innovations can emerge, develop and radiate beyond national borders.

A. African Union – Digital Transformation Strategy (2023–2030)

The African Union Digital Transformation Strategy (2023–2030) is the most comprehensive continental policy framework for digital education. It embodies an ambitious vision to make digital technology a central driver of development, economic integration, and the transformation of education systems across Africa. This strategy prioritizes schools, emphasizing the need to reduce disparities in access to technology and strengthen the capacity of education systems to integrate innovation.

One of the key objectives of this roadmap is to connect all African schools by 2030, in order to guarantee equitable access to digital educational resources. The African Union also emphasizes the promotion of local educational content, an essential dimension for ensuring the cultural and linguistic relevance of learning, while stimulating the emergence of an African digital content production industry.

The strategy also encourages the pooling of public educational platforms, with the aim of reducing costs, improving interoperability, and enabling states to share educational tools and resources in a coordinated manner. This approach seeks to avoid the fragmentation of initiatives and strengthen the coherence of national ecosystems.

Finally, the African Union is working on developing a pan-African regulatory framework dedicated to EdTech, covering essential dimensions such as technological standards, content quality, learner data protection, and accreditation mechanisms. The ambition is to create a harmonized environment conducive to the growth of African EdTech solutions and their dissemination across the continent.

Thus, the African Union's 2023–2030 strategy represents a structuring foundation for the digital transformation of African education systems, providing clear guidance and a shared vision for member states and development partners.



B. Mastercard Foundation – EdTech Fellowship

The Mastercard Foundation EdTech Fellowship is one of the most influential programs in Africa's digital education ecosystem. Designed to support the emergence of high-impact educational innovations, it annually supports cohorts of startups selected for their transformative potential, their ability to improve access to learning, and their capacity to scale. The program is currently deployed in several countries across the continent, including Senegal, Benin, Kenya, Egypt, Morocco, Rwanda, Nigeria, South Africa, Ghana, Ethiopia, and Tanzania, where it contributes to the sustainable development of local EdTech ecosystems.

The initiative relies on a network of local innovation hubs, such as Etri-Labs in Benin, Injini in South Africa, CcHub in Nigeria, iHub in Kenya, Edventure in Egypt, and MEST in Ghana. These hubs play a vital role in supporting startups by providing them with an experimental environment, access to international mentors, funding opportunities, and acceleration programs tailored to the realities of the African market. This decentralized approach allows the program to be deeply rooted in national contexts while also benefiting from a continental reach.

The Fellowship also incorporates a strong dimension of inclusion and equity, particularly regarding the participation of young women in digital entrepreneurship and the use of EdTech solutions by marginalized populations. This focus reflects the vision of the Mastercard Foundation, which places education, economic empowerment, and equal opportunities at the heart of its mission.

African Union (2020). Digital Transformation Strategy for Africa (2020–2030). Available at: https://au.int/sites/default/files/documents/38507-doc-dts-english.pdf African Union (2022). Digital Education Strategy and Implementation Plan (2023–2028). Available at: https://au.int/en/documents/20221125/digital-education-strategyand-implementation-plan UNESCO-UNEVOC (2022). Digital Education Strategy – Factsheet. Available at: https://unevoc.unesco.org/bilt/bridging-africa/1.-DES-FR-2022_09_14.pdf African Union – Official Portal. Presentation of the Digital Transformation Strategy. Available at: https://au.int/fr/node/38507

Thus, the EdTech Fellowship program not only helps to strengthen the capacity of African startups to innovate, but also to sustainably transform educational systems by promoting equity, quality and access to learning solutions adapted to the needs of the continent.

C. World Bank, Global Partnership for Education (GPE), AFD and UNICEF

International financial institutions and development agencies, including the World Bank, the Global Partnership for Education (GPE), the French Development Agency (AFD), and UNICEF, play a central role in structuring and financing the digital transformation of education systems in Africa. Their actions are part of a systemic strengthening approach, aiming to improve access to education, the quality of learning, and the effectiveness of educational governance.

For several years, these actors have supported pilot programs for the digitalization of basic education, primarily at the primary and secondary levels. These interventions target the gradual introduction of digital tools in public schools, the modernization of teaching practices, and the improvement of education ministries' capacity to plan, monitor, and evaluate educational policies using reliable data. The preferred approach often relies on controlled-scale pilot projects, allowing solutions to be tested before potential nationwide implementation.

A major focus of their work is the development of digital educational content and Education Management Information Systems (EMIS). These initiatives aim to produce educational resources aligned with national curricula, accessible both online and offline, and adapted to local contexts, including in some national languages. In parallel, investments in educational information systems contribute to strengthening governance in the sector by facilitating data collection, monitoring student performance, managing teachers, and supporting evidence-based decision-making.

These organizations are also paying increasing attention to experimenting with hybrid educational models, combining face-to-face teaching and distance learning, particularly in rural and remote areas. These models aim to address structural constraints (poor connectivity, shortages of qualified teachers, geographical isolation) while leveraging appropriate solutions such as offline-first platforms, educational radio, mobile learning, and content delivered via low-cost devices. The lessons learned from these experiments are gradually informing national and regional strategies for the digitalization of education.

Overall, the joint action of the World Bank, the GPE, the AFD, and UNICEF is helping to create an enabling environment for integrating EdTech into public policies, while highlighting the need for better integration with local innovation ecosystems. This momentum opens up significant opportunities to strengthen public-private partnerships, support solutions developed by African startups, and foster the sustainable scaling up of high-impact educational innovations.

Sources Mastercard Foundation (2023). EdTech Fellowship Program Overview. Available at : https://mastercardfdn.org/edtech-fellowship Mastercard Foundation – Center for Innovative Teaching & Learning (CITL). EdTech Ecosystem Reports & Program Descriptions. Available at : Foundation EdTech Fellowship – South Africa Implementation. Available at: https://injini.africa/edtech-fellowship

edtech-fellowship CcHub (Nigeria & Kenya). EdTech Fellowship Program Details. Available at: https://cchub.africa/programmes/edtech-fellowship

4. CONSTRAINTS AND DYNAMICS OF EDTECH FINANCING IN AFRICA

Despite significant growth potential and a recognized strategic role in transforming education systems, EdTech funding in Africa remains largely insufficient and ill-suited to the realities of the sector. This situation is one of the main obstacles to innovation, scaling up solutions, and the sustainability of African educational startups.

A. Limited access to patient capital and growth financing

Access to patient capital, essential for developing educational solutions with long adoption and monetization cycles, remains particularly limited. Data from sector reports indicate that less than 2% of venture capital investments in Africa are directed towards the EdTech sector, reflecting a persistent perception of EdTech as a low-profitability segment with high institutional dependence.
This situation is aggravated by:

- the absence of investment funds specializing in EdTech or digital education;
- the lack of public funding mechanisms dedicated to educational startups, capable of absorbing the specific risks of the sector;
- the difficulty for EdTech companies to raise funding beyond the pre-seed or seed stage, limiting their ability to invest in product improvement, user acquisition and geographic expansion.

As a result, many startups remain at an experimental stage, without any real capacity to scale up regionally or nationally.

B. Strong dependence on funding from donors

Faced with insufficient private capital, a significant proportion of African EdTech startups rely on non-recurring funding: grants, donations, and pilot projects financed by donors or foundations. While these resources play a crucial role in the initial phases of innovation and prototyping, they nevertheless have structural limitations.
This dependency translates into:

- low financial predictability, making medium and long-term planning difficult;
- a high risk of project discontinuity at the end of funding cycles;
- an sometimes excessive focus on the priorities of donors, to the detriment of economic viability and market suitability.

Furthermore, hybrid economic models combining, for example, B2G (Business to Government) and B2B (Business to Business) approaches remain underdeveloped and insufficiently supported, even though they could constitute a key lever for sustainability in the sector.

C. Emergence of new financing opportunities

Despite these constraints, several positive signs indicate a gradual evolution of the EdTech financial landscape in Africa. These include:

- the rise of hybrid impact funds, specializing in education and skills, such as Acumen, Imaginable Futures or other mechanisms combining financial return and social impact;

- a growing interest in structuring public-private financing arrangements, including sovereign wealth funds, state guarantee mechanisms or risk-sharing instruments;

- the still largely untapped potential of public education procurement, which could become a major growth driver for local EdTech solutions if suitable and transparent procurement frameworks were put in place.

From this perspective, the development of intermediation platforms, specialized hubs and innovative financing mechanisms appears as an essential condition for aligning capital, public policies and educational innovation, and thus enabling African EdTech to move from experimentation to a systemic and sustainable impact.

5. STRATEGIC RECOMMENDATIONS

To foster the sustainable, inclusive, and sovereign development of EdTech in Africa, African states, in partnership with regional organizations, donors, the private sector, and innovation stakeholders, are urged to adopt a structured, coordinated, and evidence-based approach. The following recommendations aim to strengthen the impact of educational technologies while ensuring their alignment with national and continental priorities.

A. Develop and institutionalize national EdTech frameworks

Governments should define and formalize national frameworks dedicated to EdTech, integrated into existing educational and digital policies. These frameworks should:

- To be aligned with national curricula, local realities and priorities in terms of employability and skills of the future;
- Provide specific budget lines within national education and digital budgets, in order to reduce dependence on one-off funding;
- To be broken down into operational action plans by educational level (basic education, secondary education, vocational training, higher education), with clear objectives, measurable indicators and defined responsibilities.

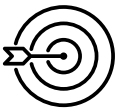
Such an approach would make it possible to move from fragmented initiatives to a systemic and coherent integration of EdTech into educational systems.

B. Create sovereign educational innovation funds

To address the patient capital shortage in the sector, it is recommended to establish sovereign educational innovation funds specifically dedicated to EdTech. These funds could:



to be funded by a combination of national, multilateral and philanthropic resources, in order to pool the risks;



prioritize patient funding for local EdTech startups, particularly in the seed and growth stages;



be associated with rigorous impact monitoring and evaluation mechanisms, covering educational outcomes, equity of access and economic viability.



These instruments contribute to strengthening educational sovereignty and supporting the emergence of endogenous solutions adapted to African contexts.

C. Develop structured public-private partnerships that promote innovation

States should promote structured public-private partnerships (PPPs), designed as levers for innovation and scaling up. This notably implies:

- the implementation of public tenders open to local innovation, with criteria adapted to young companies and emerging solutions;
- the creation of national or regional platforms for the evaluation, approval and labelling of EdTech solutions, guaranteeing their educational quality, safety and regulatory compliance;
- the explicit integration of EdTech solutions into policies for school connectivity, digital inclusion and the transformation of public educational services.

These mechanisms would strengthen the confidence of public institutions in local solutions and facilitate their widespread adoption.

D. Strengthen cross-sectoral governance

The success of EdTech strategies depends on strengthened cross-sectoral governance that transcends institutional silos. States are encouraged to:

improve coordination between the ministries in charge of Education, Digital Affairs, Finance and Vocational Training

define and adopt common standards for system interoperability, data protection, cybersecurity and educational quality

promote regular consultation frameworks involving public actors, the private sector, startups, universities and civil society.

Integrated governance is essential to ensure consistency of interventions and maximize the impact of investments.

E. Support the production and use of evidence

Finally, it is crucial to strengthen the production, analysis, and dissemination of evidence on EdTech. Efforts should focus on:

- measuring the real impact of EdTech solutions on learning, school retention, equity and inclusion;
- the development of harmonized monitoring and evaluation tools, enabling national and regional comparisons;
- open and transparent dissemination of results, in order to inform policy decisions, guide investments and promote collective learning.

Anchoring EdTech policies in reliable and shared data is a key condition for a sustainable and evidence-based educational transformation.

SECTION 5

TRENDS, CHALLENGES AND FUTURE PROSPECTS FOR EDTECH IN AFRICA

1. EMERGING TRENDS IN AFRICAN EDTECH

The African EdTech ecosystem, still young, is in a phase of acceleration, driven by converging regional dynamics, rapid urbanization, demographic pressure, increasing mobile penetration and a massive demand for alternative educational solutions.

A. Hybridization of Pedagogical Models: A Key Trend in African EdTech

The hybridization of educational models represents a major transformation, aiming to optimize teaching and learning by leveraging the strengths of both digital and in-person learning. This approach is particularly relevant in the African context, which faces challenges related to access, quality, and the expansion of education.

1. **Proliferation of Blended Learning Formats (In-Person + Remote)**

Blended learning is at the heart of this hybrid approach. It's not about replacing traditional teaching, but about enriching it and making it more flexible.

- **Enriching the Classroom Experience**



Class time (in-person) is now more dedicated to collaborative activities, discussions, mentoring, and solving complex problems. Theory and the acquisition of basic knowledge are often transferred to home or outside of class hours, via digital means.

- **Flexibility and accessibility**



Distance learning allows students in rural or remote areas, or those with professional or family constraints, to access quality content. This promotes differentiated instruction, where each student can progress at their own pace.

The Flipped Classroom Model: A concrete example of blended learning where lectures are viewed at home (distance learning) and class time is used for exercises and application (face-to-face).

2. Integration of LMS Platforms in Schools and Universities

Learning Management Systems (LMS), or learning management platforms (such as Moodle, Canvas, or local solutions), are becoming essential tools for structuring hybrid learning.

- **Centralization of Content: LMSs allow for the centralized storage and organization of course materials, videos, quizzes, assignments, and additional resources.**
- **Educational monitoring and assessment: They offer teachers the opportunity to track student progress in real time, administer online assessments, and provide personalized and instant feedback.**
- **Interaction and Collaboration: These platforms often integrate discussion forums, chat tools and collaborative workspaces that extend educational interaction beyond the classroom.**

3. Growing Coexistence between Formal, Informal and Non-Formal Education

Learning Management Systems (LMS), or learning management platforms (such as Moodle, Canvas, or local solutions), are becoming essential tools for structuring hybrid learning.

Type of Education	Definition	Role of African EdTech
Formal	Structured education system (schools, universities).	Integration of LMS, accredited online courses (MOOCs), digital resources.
Non-formal	Learning organized outside the formal system (vocational training, certification workshops).	Micro-credential platforms, tutorials for specific skills (coding, entrepreneurship), short certifications
Informal	Continuous learning from everyday experiences (reading, social interactions, web).	Open content platforms (YouTube, educational blogs), self-learning tools, online communities of practice.

This coexistence allows African learners to acquire specific and immediately employable skills, often more quickly and at a lower cost than through purely formal means, while also valuing the knowledge and experiences acquired outside the traditional school curriculum.

B. Growth of Contextualized Content: Promoting Inclusivity and Cultural Resonance

Simply importing digital educational resources from the West or other regions is ineffective. For EdTech to have a significant impact, content must speak directly to African learners, reflect their reality, and use their language.

1. Growth of Educational Content in Local Languages

Education in a language a child does not master is a major obstacle to learning. Creating content in local languages is essential for access and comprehension.

- Learning fundamental concepts (mathematics, science) in one's native language or a known local language greatly facilitates assimilation and reduces academic failure, especially at the primary level.
- The use of local languages in EdTech media helps to preserve and enhance the African linguistic heritage, which is immensely diverse.
- Technologies (machine translation, multilingual speech recognition, translation crowdsourcing platforms) are increasingly being used by local startups to quickly adapt existing content.



2. Development of Resources Adapted to Sociocultural Contexts

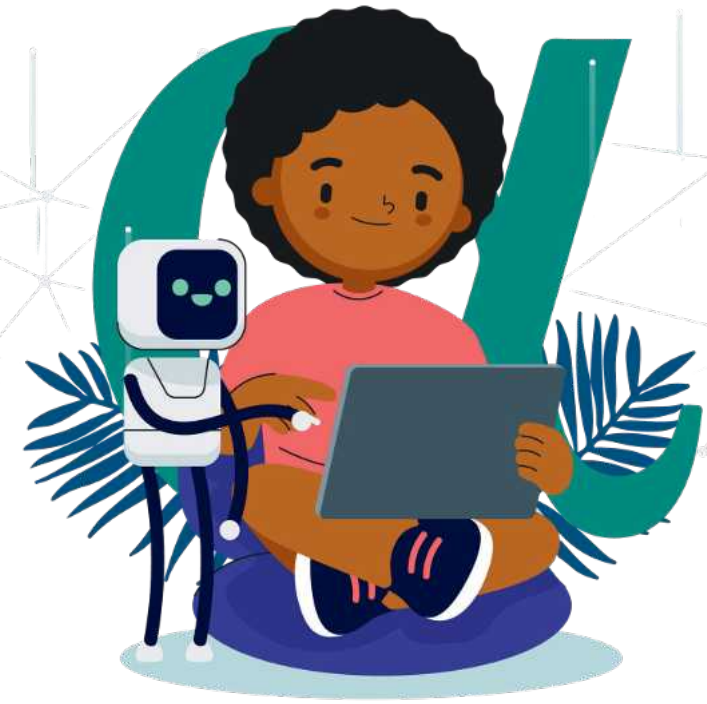
Contextualization goes beyond simple translation; it involves integrating local realities into learning scenarios.

- Math exercises should use local currencies, physics problems should refer to African infrastructure (roads, power grids), and case studies should concern African entrepreneurs or socio-economic challenges.
- The content should incorporate the stories, myths, community values and historical figures of the continent, thereby strengthening learners' identity and motivation.
- Learning modules on topics such as sustainable agriculture, water management, solar energy or public health challenges (e.g., the fight against malaria) are much more relevant and engaging when they are rooted in the local context.



3. Emergence of Digital Educational Media

Using engaging and modern formats is crucial to capturing the attention of young learners.



- **Educational Cartoons: These formats are excellent for teaching the basics of reading, arithmetic, or hygiene rules in a fun and memorable way. Local productions are emerging, featuring characters and environments familiar to African children.**
- **Serious Games: These transform learning into interactive challenges. For example, a game might simulate managing a small farm business or solving a local environmental problem, allowing learners to develop practical skills through risk-free experimentation.**
- **Local Audio and Video Content: With the increasing penetration of smartphones, short explainer videos and podcasts produced by local experts or educators are becoming powerful tools for mobile learning (m-learning).**

Conclusion of the Approach: This trend towards contextualized content not only promotes inclusivity (by lowering linguistic and cultural barriers), but it also ensures cultural resonance, making education more meaningful and effective for the African learner.

4. Increased Power of Alternative Certification

In a continent where unemployment among young graduates is a major challenge, alternative certification offers a fast and relevant path to employability.

- **Growth of Micro-certifications, Digital Credentials, Badges:** These digital formats attest to the acquisition of a specific and measurable skill (e.g., knowing how to code in Python, mastering data analysis, or managing a supply chain for SMEs). They are often awarded upon completion of short, intensive courses offered by EdTech platforms.
- **Progressive Recognition by Employers:** Companies, often frustrated by the gap between traditional university degrees and immediate market needs, are beginning to place more value on this evidence of concrete and up-to-date skills. This creates a direct link between digital training and employment.
- **Partnerships between Startups and Universities to Validate Courses:** To gain legitimacy, many EdTech companies are establishing partnerships with recognized higher education institutions. These collaborations allow learners to obtain academically validated certificates while benefiting from the flexibility and relevance of online courses.

Result: Alternative certification is a direct response to the gap between traditional degrees and the skills required in the African job market, promoting a more agile and employable workforce.

E. Growing Place of AI and Learning Analytics

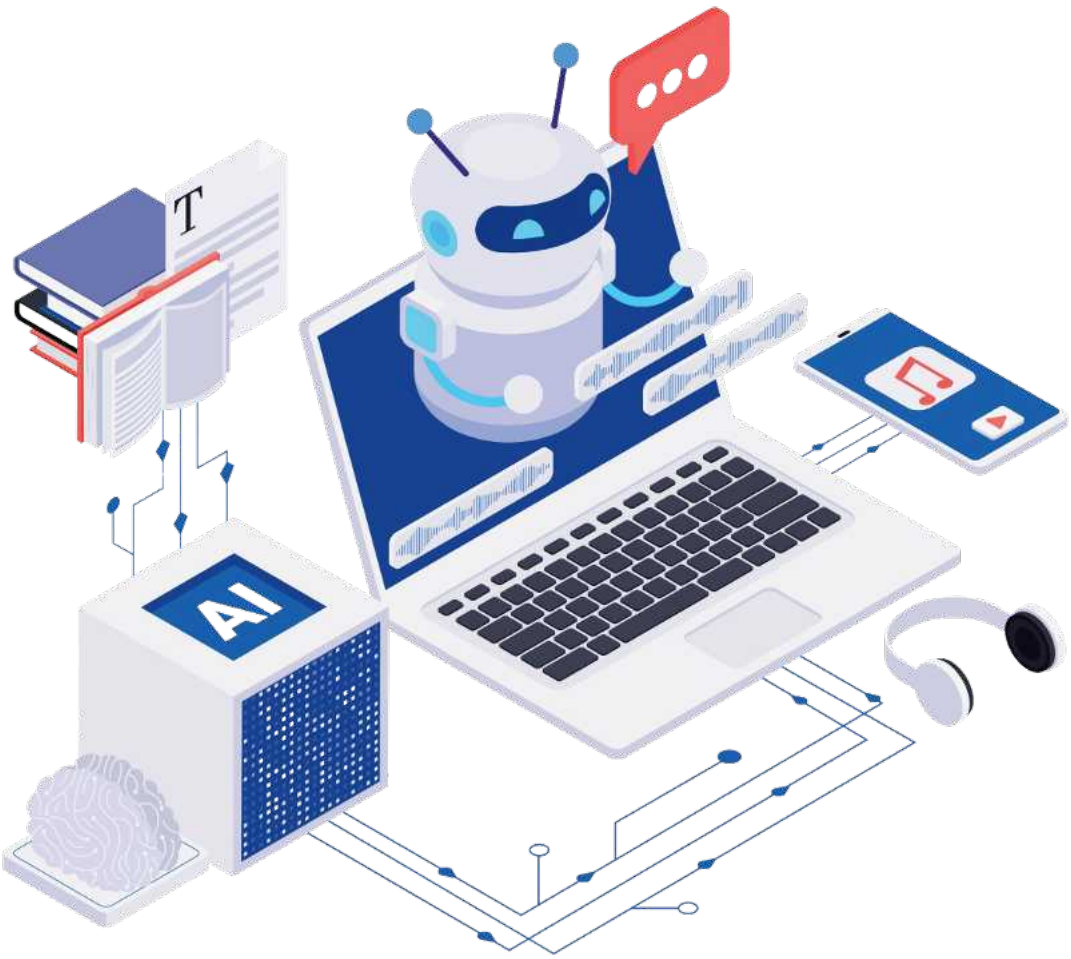
The integration of advanced technologies such as Artificial Intelligence (AI) and learning analytics is the next horizon for African EdTech.

First AI Integrations:

Adaptive Tutoring: AI is used to personalize each student's learning path, adjusting the difficulty of exercises and content based on their strengths and weaknesses, thus enabling individualized learning on a large scale.

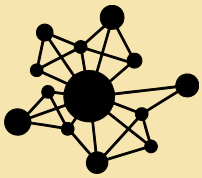
- **Automated Assessment:** AI helps to instantly correct certain types of work or to quickly identify areas of difficulty for teachers.
- **Development of Real-Time Learning Tracking Tools:** Learning analytics collects data on how learners interact with platforms (time spent, success rates, recurring errors). This data allows teachers and administrators to manage educational performance and identify students at risk of dropping out.
- **Use of Data to Guide Educational Policies:** At a macro level, the aggregation of learning data can inform governments about the effectiveness of curricula, missing skills in certain regions, and direct investments towards the most critical areas.

Result: AI and learning analytics represent a new technological frontier with strong potential for optimizing the quality of education in Africa. However, their adoption is still hampered by infrastructure challenges and the need to train educators to use these complex tools.



2. Persistent Challenges and Risks to Anticipate

Despite positive trends (A to E), EdTech in Africa faces major structural and cyclical obstacles that threaten the sustainability and impact of these innovations.



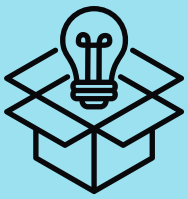
Ecosystem Fragmentation

The absence of unified national regulatory frameworks and the proliferation of non-interoperable initiatives lead to a risk of wasted resources and duplication of effort. The lack of platform and data sharing prevents the creation of the critical mass necessary for systemic adoption.



Fragility of Economic Models

Low household purchasing power and very limited access to patient capital (which tolerates a slow return on investment) make B2C (direct-to-consumer) and B2G (business-to-government) models difficult to scale. Startups struggle to survive beyond the pilot phase without structured support mechanisms.



Gap between Innovation and the Education System

EdTech innovation often remains marginal. Integration into school curricula is weak, and institutional mistrust of private solutions persists. Furthermore, resistance to change from some teaching staff hinders adoption, rendering innovation less impactful if it is not adopted by the formal education system.



Limited Digital Skills

The challenge is twofold:
Infrastructure: Insufficient connectivity and equipment inadequate in many schools.
Human capital: Low digital skills among many teachers and decision-makers.



Lack of Evidence of Impact

The lack of rigorous documentation and shared evaluation protocols makes it difficult to establish the true effectiveness of solutions. This absence of measurable results (proof of impact) slows institutional investment and the ability to convince funders of the value of maintaining funding.

APPENDICES

1. Acknowledgments

We express our gratitude to the United Nations Development Programme (UNDP) for the trust placed in the completion of this strategic study, carried out within the framework of the timbuktoo programme and the process of setting up the Dakar EdTech Hub.
Our thanks go to our partners and national institutions for their crucial support, in particular:

the Ministry of National Education of Senegal, through SENUM;
the Ministry of Higher Education, Research and Innovation;
Cheikh Hamidou Kane University (UNCHK);
as well as partner universities, higher education and vocational training institutions.

We also thank the public and semi-public structures, tech hubs, incubators and innovation support programs for their institutional contribution and commitment.

Finally, we acknowledge the support of the timbuktoo program coordination teams, whose technical assistance and strategic alignment were essential to the conduct of this study.

2. List of stakeholders consulted

This study and the strategic roadmap for the Dakar EdTech Hub were developed based on a broad and inclusive consultation process involving key stakeholders in the Senegalese and regional EdTech ecosystem. The stakeholders consulted include, but are not limited to:

Public institutions

- Ministry of National Education
- Ministry of Higher Education, Research and Innovation
- Ministry of Vocational and Technical Training
- Ministry of Digital Economy
- Public agencies and affiliated structures

Universities and training institutions

- Public and private universities
- Higher education institutions
- Vocational and technical training centers

EdTech Startups and Companies

- EdTech startups in the seed, growth, and scaling phases
- Providers of digital educational solutions

Tech hubs, incubators and accelerators

- University incubators (UNIPOD)
- Innovation hubs and technology accelerators

Technical and financial partners

- International organizations
- and development agencies
- Foundations and impact investors

Independent experts

- Specialists in pedagogy,
- educational technologies,
- public policies
- and innovation funding

The detailed list of organizations and institutions consulted is available on the dynamic mapping platform: https://edtechecosystem.org/

3. Methodological tools used

The study is based on a mixed-methods approach, combining document analysis, primary data collection, and participatory co-construction. The main tools used include:

In-depth literature review (institutional reports, sector studies, national and regional policies)

Structured questionnaires addressed to EdTech startups, hubs and support programs. Semi-structured interviews with public decision-makers, university officials, entrepreneurs and financial partners. Co-construction and strategic validation workshops with ecosystem stakeholders.
Tools for mapping actors, needs and EdTech solutions. Analytical frameworks inspired by the approaches of UNDP, the World Bank, the EdTech Hub and HolonIC. This methodology has ensured the analytical robustness, representativeness of viewpoints and operational anchoring of recommendations.



4. Bibliographical References

The report's analyses and recommendations are based on internationally recognized sources, including: United Nations Development Programme (UNDP); UNESCO Timbuktu Programme Documents; World Bank Global Education Monitoring Report; UNICEF Education and Digital Transformation Reports; EdTech Hub Digital Learning and Connectivity in Education; HolonIQ Evidence and Country Scoping Studies; GSMA Africa EdTech Market Intelligence; Mobile Connectivity and Digital Inclusion Reports; WAEMU regional studies and national analyses (Senegal).

5. Key players by country

SENEGAL

- Xarala → Upskilling
- Yello App → K-12
- Tooshare → Marketplace
- Databeez → AI / Data
- Bakeli School of Technology → Upskilling
- Neolean Academy → Upskilling
- Ashita-ICC → TVET
- Jangalma → TVET
- Cashew → K-12
- Steamtastic → K-12 (STEM)
- 20/20 EdTech → Platform
- Education Channel → Content
- Digital School Technologies → LMS
- Edacy → Upskilling
- Grow Academy → Upskilling
- Mjangale → K-12
- Meelam → Contents
- Volkeno → TVET / Upskilling

BENIN

- LOGECO → TVET
- TAMA → K-12
- IRAWO → Contents
- Ecolia → K-12
- Kankode → Upskilling
- Malam → K-12
- Mobidoto → Marketplace
- Steamy → K-12 (STEM)
- Pocket Tutor → Marketplace
- Problems → K-12
- Epitech Technology → Upskilling
- Ilewemi → K-12
- Mara Academy → Upskilling
- Seven → K-12

IVORY COAST

- BACprep → K-12
- Bibliosave → Contents
- Chalkboard Education → LMS
- Coach'ON → Upskilling
- Dictable → K-12
- Educate → LMS
- EducforMooc → Higher Ed
- Edudesk → LMS
- Edugate → Platforms
- Eduweb → Platform
- Expand Education → K-12
- Enko Education → K-12
- Fatala Digital House → Upskilling
- Gomycode → Upskilling
- EdTech CASE → Platform
- Nan Academy → Upskilling
- Nimba Edition → Contents
- Qelasy → LMS
- TOOTREE CI → K-12
- Trace Academia → Upskilling / Contents

MALI

- RobotsMali (AI4D Lab) → AI / Upskilling
- Kabakoo Academies → Upskilling

BURKINA FASO

- Emera AI → AI
- DigiClass → LMS
- Faso e-Education → Platform

TOGO

- ANFCT e-Learning → Platform
- Suisco → LMS
- Zonoa → Marketplace

NIGER

- CET ASNI → TVET
- Codeloccol → Upskilling
- Karatou Post-Bac → Higher Ed

GUINEA

- CET ASNI → TVET
- Codeloccol → Upskilling
- Karatou Post-Bac → Higher Ed

GHANA

- Dext Technology → IA
- Ayaprep → K-12
- QKnow EdTech → K-12
- Wonderspaced →
- DeafCanTalk → AI / Inclusion
- Firefly IO → Ups
- Coral Reef Innovation Hub → Upskilling

KENYA

- University of Nairobi → Higher Ed
- Aga Khan University → Higher Ed
- Expand Education → K-12

CHAD

- Genoskul → K-12
- ChadEducationPlus → Platform

GAMBIA

- EduCare → K-12
- Leap Learning → K-12

NIGERIA

- DoLessons → Marketplace
- Tuteria → Marketplace
- SchoolsFocus → LMS
- Afribary → Higher Ed
- uLesson → K-12
- Teesas → LMS
- Gradely → K-12
- Nigenius → K-12
- Prepclass → K-12
- Edusko → Marketplace
- Traindemy → Upskilling
- ScholarX → Higher Ed

CAMEROON

- Sweetch → K-12
- Studirium → Higher Ed
- Maya Academy → Upskilling
- OSIA Technologies → LMS

CONGO

- Mashidano → K-12
- Schoolap → LMS
- Congo-Education → Platform

MOROCCO

- Yool.education → K-12
- Kezakoo → K-12
- Prove → Upskilling
- NOBOX LAB → Upskilling

EGYPT

- Sprints → Upskilling
- El Kheta → K-12
- TeacherKit → LMS
- Crafty Workshop → TVET
- iSchool → K-12
- ShareEdu → Platform
- Nafham → K-12
- Egypt Education Platform → Platform

ALGERIA

- LabLabee → Upskilling
- Esaboura → K-12
- Edutech.dz → Platform
- Gomycode Algeria → Upskilling
- IFRISQ e-Learning → LMS

TUNISIA

- ClassQuiz → K-12
- OBM Education → TVET
- Atom Academy → K-12
- School Upgrader → LMS

ANGOLA

- Rosco Edutec → K-12
- Mitrelli Education → Higher Ed
- Grade → K-12

ZAMBIA

- Colossus App → K-12
- Omni Learning → K-12
- Simba Education → K-12

SOUTH AFRICA

- Ambani Africa → Upskilling
- Book Village → Marketplace
- GetSmarter → Upskilling
- Snapplify → Content
- WeThinkCode → Upskilling
- Zaio → Upskilling
- FoondaMate → K-12
- Reflective Learning → Content
- Valenture Institute → Higher Ed

GABON

- Sikul → K-12
- Prepdia → Higher Ed
- ESTIM Gabon → Higher Ed
- ESIITECH → Upskilling

RWANDA

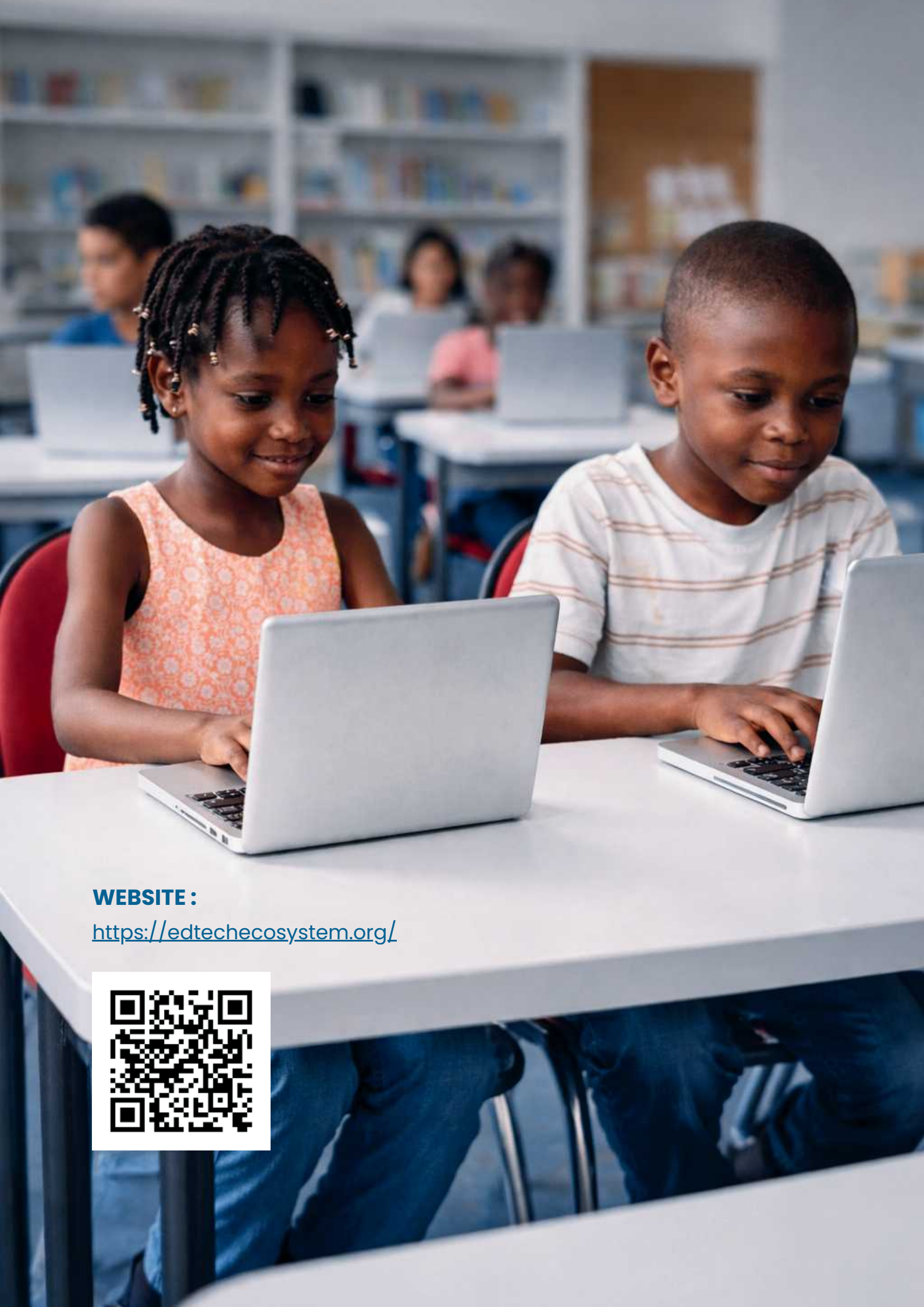
- AltSchool Africa → K-12
- ALX → Upskilling
- SOLVIT AFRICA → Upskilling
- Cyber University / SoftBank → Higher Ed
- O'Genius → Upskilling
- Rwanda Coding Academy → TVET

LIBERIA

- CEATECH-SL → TVET
- YONER Liberia → K-12
- AIIST → Higher Ed
- Weeglo → K-12

6. Glossary of EdTech Terms

- **EdTech (Educational Technology):** The set of technologies, platforms and digital solutions applied to education and training.
- **Blended Learning:** A teaching model combining face-to-face teaching and distance learning.
- **E-learning:** A learning method based primarily on digital platforms accessible online or offline.
- **M-learning (Mobile Learning):** Learning using mobile devices such as phones and tablets.
- **EMIS (Education Management Information System):** Information system for the collection, processing and analysis of educational data to support governance.
- **OER (Open Educational Resources):** Open, freely accessible and reusable educational resources, including courses, manuals and teaching materials.
- **Learning Management System (LMS):** A learning management platform that enables content distribution, learner tracking, and evaluation.
- **Artificial intelligence (AI) in education:** Application of AI techniques to personalize learning, automate assessment and improve teaching effectiveness.
- **Blended finance (hybrid financing):** Mechanisms combining public, private and philanthropic financing in order to reduce risks and attract private capital.
- **Patient capital:** Long-term financing that prioritizes social impact and sustainability over immediate return.
- **EdTech startup:** A young company developing innovative technological solutions to meet educational needs. Venture building: A structured approach to supporting the creation and growth of startups.
- **Evidence-based policy:** The development of public policies based on empirical data and rigorous evaluations.
- **Regulatory sandbox:** An experimental framework for testing innovations in a controlled regulatory environment.
- **Gamification:** Integration of game mechanics into learning systems to enhance learner engagement.
- **Interoperability:** The ability of different digital systems and platforms to work together and exchange data.
- **Educational digital sovereignty:** The ability of a country or region to control its digital educational infrastructure, data, and content.
- **UniPod (University Innovation Pod):** A university incubator supported by UNDP aimed at stimulating innovation and entrepreneurship.



WEBSITE :

https://edtechecosystem.org/